Chapter 1 — The Mathematical Foundations

This chapter is a **pre-requisite mathematical foundation** for the generative modeling chapters that follow.

Learning Objectives

By the end of this chapter, you should be able to:

1. Explain **random variables and expectations**
2. Understand **rules of expectations**
3. Know the difference between **conditional** and **marginal** distributions
4. Understand **Jensen’s inequality**
5. Manipulate the relationships between **joint**, **conditional** and **marginal** distributions
6. Understand **Law of Total Expectations** and **Law of Variance**
7. Understand the relationship between **Kullback–Leibler (KL) Divergence**, **Entropy**, and **Cross-Entropy** and **Mutual Information**
8. Know the concept of **Maximum Likelihood Estimations**
9. Know how **Monter Carlo Estimation** works

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In [25]:

```python
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.dpi"] = 120
np.random.seed(0)
```

1. Random Variables and Expectation

Let X be a random variable.

Discrete

$\mathbb{E}[X] = \sum_x x\,p(x)$

Continuous

$\mathbb{E}[X] = \int x\,p(x)\,dx$

More generally, for a function f:

$$\mathbb{E}[f(X)] = \sum_x f(x)\,p(x) \quad \text{or} \quad \int f(x)\,p(x)\,dx$$

Interpretation: expectation is a **weighted average** value of a function under the distribution.

In [ ]:

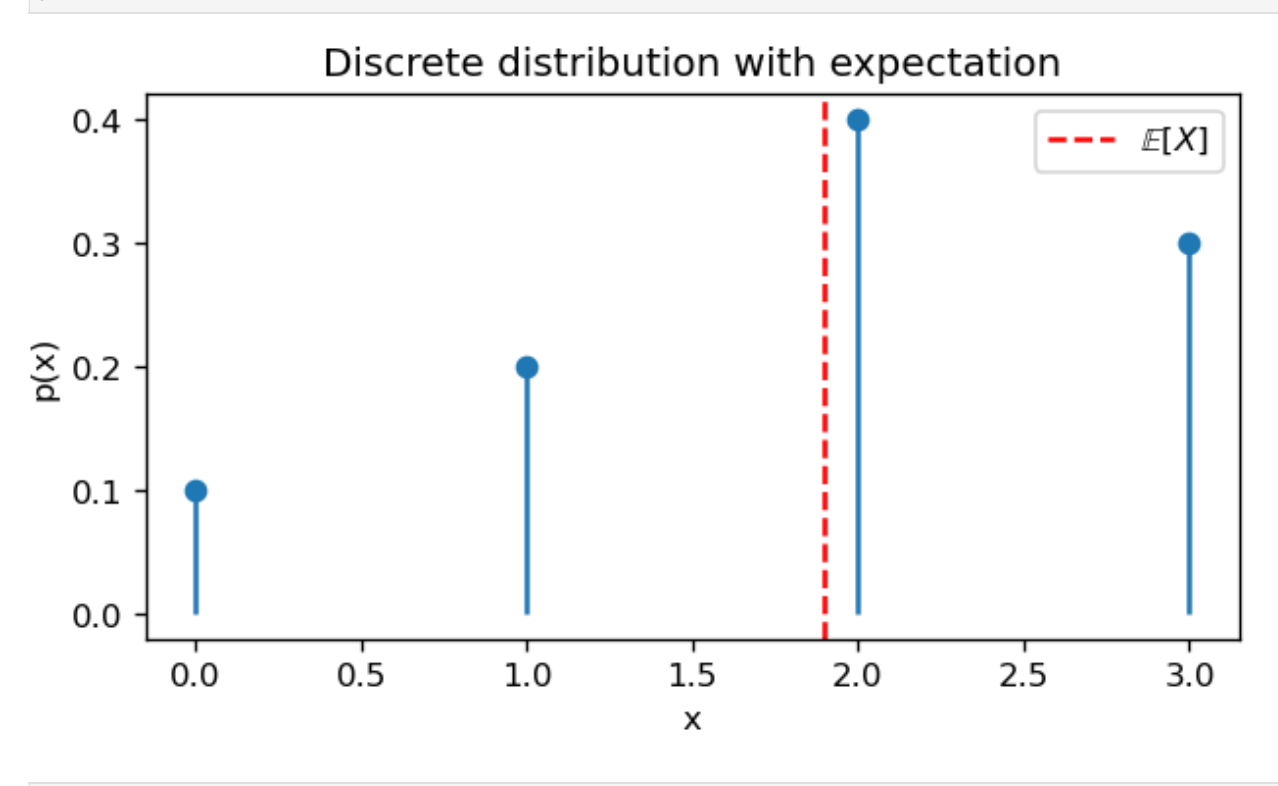
```python
# Discrete example
x = np.array([0, 1, 2, 3], dtype=float)
p = np.array([0.1, 0.2, 0.4, 0.3], dtype=float)

# make sure it is a true probability distribution
p = p / p.sum()

E_X  = np.sum(x * p)
E_X2 = np.sum((x**2) * p)

# note that expected values is just a number

E_X, E_X2
```

Out[ ]:

```
(1.9, 4.5)
```

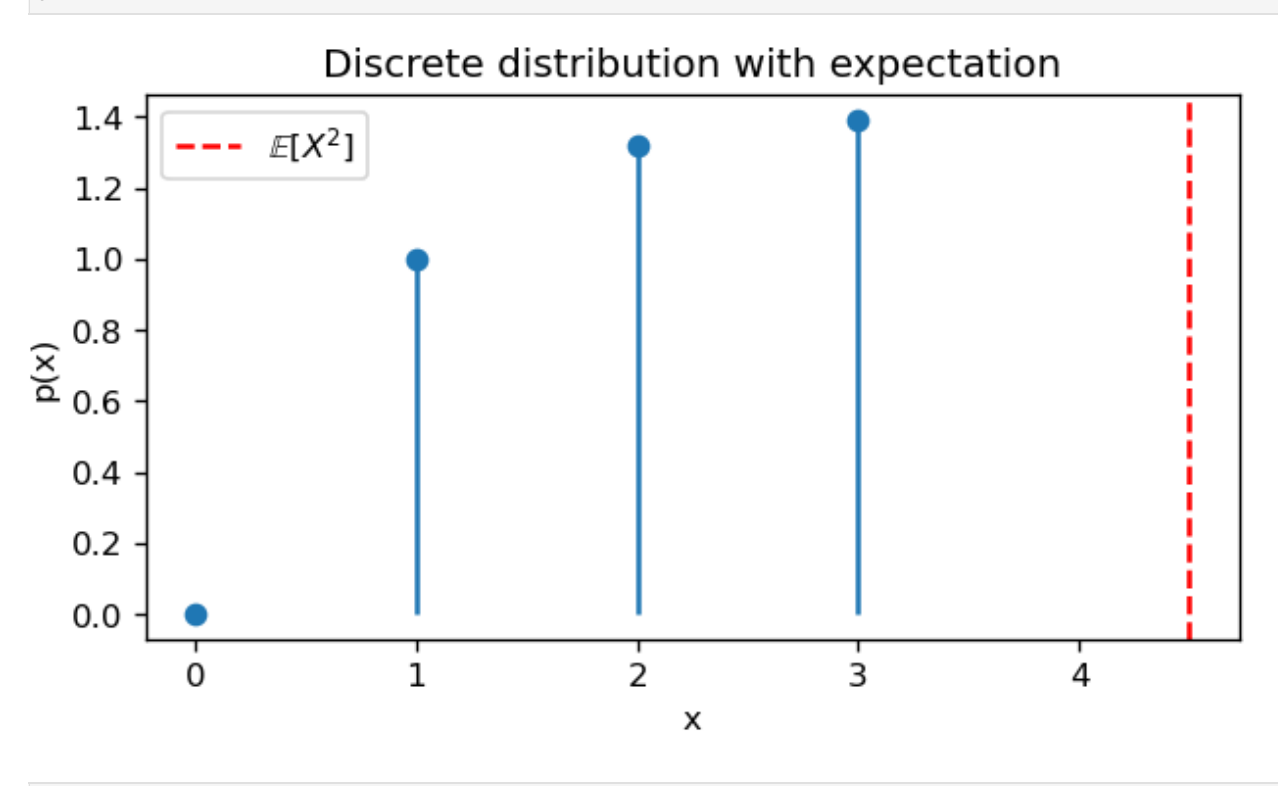
In [27]:

```python
fig, ax = plt.subplots(figsize=(6,3))

ax.stem(x, p, basefmt=" ")
ax.axvline(E_X, color="red", linestyle="--", label=r"$\mathbb{E}[X]$")

ax.set_xlabel("x")
ax.set_ylabel("p(x)")
ax.set_title("Discrete distribution with expectation")
ax.legend()

plt.show()
```



In [28]:

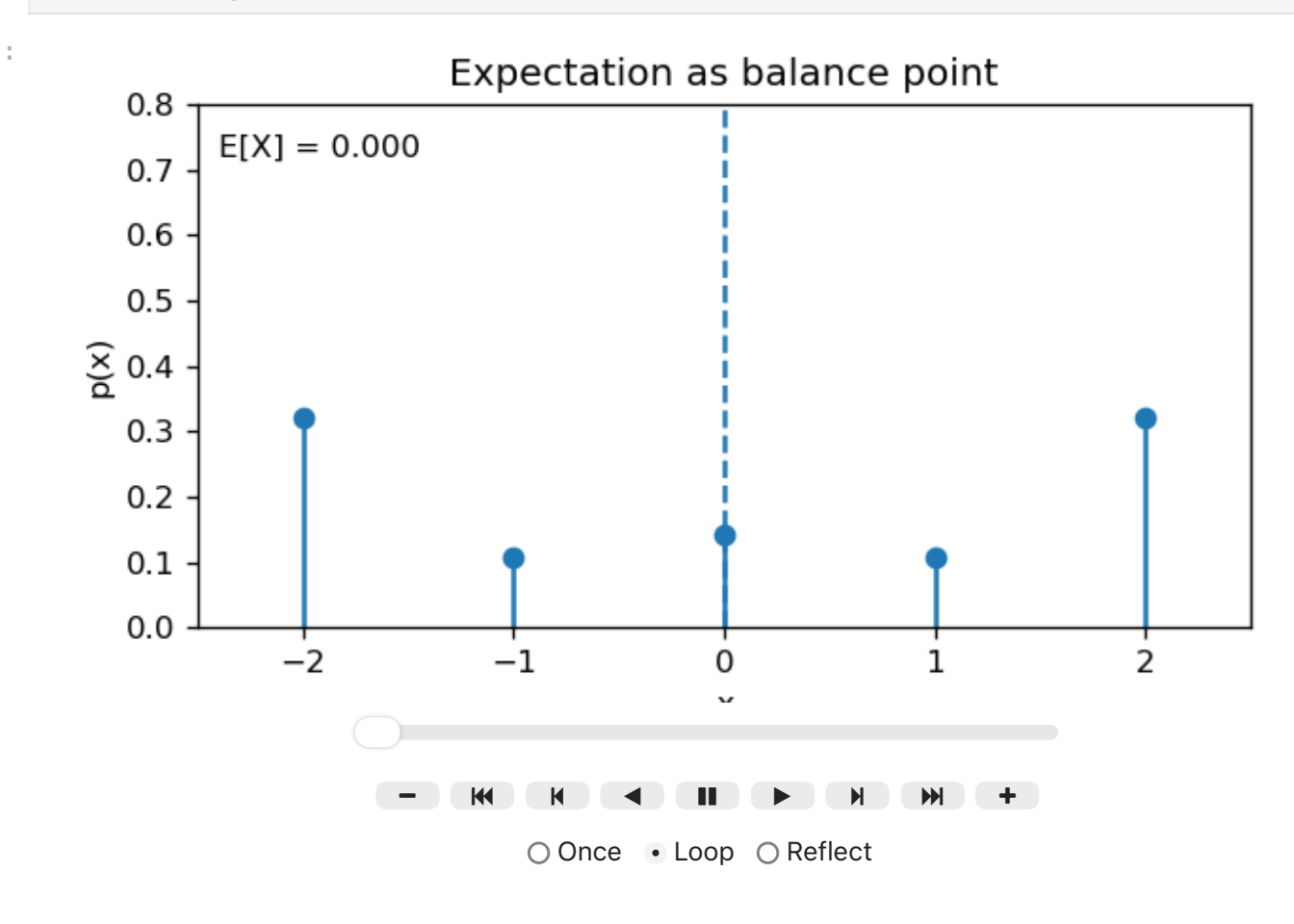
```python
fig, ax = plt.subplots(figsize=(6,3))

ax.stem(x, x**p, basefmt=" ")
ax.axvline(E_X2, color="red", linestyle="--", label=r"$\mathbb{E}[X^2]$")

ax.set_xlabel("x")
ax.set_ylabel("p(x)")
ax.set_title("Discrete distribution with expectation")
ax.legend()

plt.show()
```



In [29]:

```python
from matplotlib.animation import FuncAnimation
from IPython.display import HTML

x = np.array([-2, -1, 0, 1, 2], dtype=float)
fig, ax = plt.subplots(figsize=(6,3))
ax.set_xlim(-2.5, 2.5)
ax.set_ylim(0, 0.8)
ax.set_xlabel("x")
ax.set_ylabel("p(x)")
ax.set_title("Expectation as balance point")

stem = ax.stem(x, np.zeros_like(x), basefmt=" ")
exp_line = ax.axvline(0, linestyle="--")
exp_text = ax.text(-2.4, 0.72, "")

def frame(t):
    w = (np.sin(2*np.pi*t/80) + 1)/2
    p = np.array([0.45*(1-w), 0.15*(1-w), 0.10, 0.15*w, 0.45*w], dtype=float)
    p = p/p.sum()
    E = np.sum(x*p)

    markerline, stemlines, baseline = stem
    markerline.set_ydata(p)
    stemlines.set_segments([[(xi, 0), (xi, pi)] for xi, pi in zip(x, p)])
    exp_line.set_xdata([E, E])
    exp_text.set_text(f"E[X] = {E:.3f}")
    return markerline, stemlines, exp_line, exp_text

anim = FuncAnimation(fig, frame, frames=80, interval=60, blit=False)
plt.close(fig)
HTML(anim.to_jshtml())
```

Out[29]:



2. Rules of Expectation (Linearity)

For constants a, b, c:

$$\mathbb{E}[aX + bY + c] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y] + c$$

Holds even if X and Y are dependent.

Other immediate consequences:

$$\mathbb{E}\left[\sum_i X_i\right] = \sum_i \mathbb{E}[X_i]$$

$$\mathbb{E}[c] = c$$

Testing Empirically

We can use a monte carlo experiment to generate random numbers for X and Y and show that the equation above holds.

In [ ]:

```python
# Monte Carlo check of linearity (including dependence)
N = 200_000

# Create an X based on standard normal
X = np.random.normal(loc=0.0, scale=1.0, size=N)

# Create a Y that is correlated with X
Y = 0.7*X + np.random.normal(loc=1.0, scale=1.5, size=N)  # correlated with X

# mean representation of linearity of expectation
lhs = np.mean(2*X + 3*Y + 5)
rhs = 2*np.mean(X) + 3*np.mean(Y) + 5

# note that they are almost the same and the error is very small
(lhs, rhs, abs(lhs-rhs))
```

Out[ ]:

```
(8.020601919467802, 8.020601919467804, 1.7763568394002505e-15)
```

3. Conditional Expectation

$$\mathbb{E}[X \mid Y = y] = \sum_x x\,p(x \mid y)$$

Key point:

$$\mathbb{E}[X \mid Y]$$

is a random variable (a function of Y).

Simple discrete visual example:

Assume that X takes discrete values $\{1, 2, 3, 4\}$ and Y is **binary**, with $Y \in \{0, 1\}$. We also assume a joint distribution $p(x, y)$ given explicitly in a table.

The conditional expectation is computed as:

$$\mathbb{E}[X \mid Y = y] = \sum_x x\,p(x \mid y)$$

Visually:

- Fix a value of Y
- Look only at the column corresponding to that Y
- Take the **weighted average** of the X values in that column

In [35]:

```python
import numpy as np
import matplotlib.pyplot as plt

# Define discrete X and Y
X_vals = np.array([1, 2, 3, 4])
Y_vals = np.array([0, 1])

# Joint distribution p(x,y)
# rows: X = 1,2,3,4
# cols: Y = 0,1
p_xy = np.array([
    [0.20, 0.05],
    [0.25, 0.10],
    [0.10, 0.25],
    [0.05, 0.00]
])


# Sanity check probabilities should sum to 1
assert np.isclose(p_xy.sum(), 1.0)

# Marginals sum across columns for p(y|x)
p_y = p_xy.sum(axis=0)
print("p(y=0):", p_y[0])
print("p(y=1):", p_y[1])
p_x_given_y = p_xy / p_y
print("p(x|y=0):", p_x_given_y[:,0])
print("p(x|y=1):", p_x_given_y[:,1])

# Conditional expectations where @ is matrix multiplication
E_X_given_Y = X_vals @ p_x_given_y

p_y, E_X_given_Y
print("E_X_given_y0:", E_X_given_Y[0])
print("E_X_given_y1:", E_X_given_Y[1])
```

```
p(y=0): 0.6000000000000001
p(y=1): 0.4
p(x|y=0): [0.33333333 0.41666667 0.16666667 0.08333333]
p(x|y=1): [0.125 0.25  0.625 0.   ]
E_X_given_y0: 1.9999999999999998
E_X_given_y1: 2.5
```

Table view of the joint distribution

Each column is a conditional distribution $p(x \mid Y = y)$ after normalization.

In [36]:

```python
# Display joint and conditional distributions
print("Joint distribution p(x,y):")
print(p_xy)

print("\nMarginal p(y):")
print(p_y)

print("\nConditional distributions p(x|y):")
for j, y in enumerate(Y_vals):
    print(f"Y={y}: p(x|y) =", p_x_given_y[:,j])
```

```
Joint distribution p(x,y):
[[0.2  0.05]
 [0.25 0.1 ]
 [0.1  0.25]
 [0.05 0.  ]]

Marginal p(y):
[0.6 0.4]

Conditional distributions p(x|y):
Y=0: p(x|y) = [0.33333333 0.41666667 0.16666667 0.08333333]
Y=1: p(x|y) = [0.125 0.25  0.625 0.   ]
```

This means that p(y=0) = 0.6 across all values of x, and p(y=1)=0.4 across all values of x. Similarly, $p(x \mid y)$ is normalizing $p(x, y)$ by $p(y)$

Visual: conditional expectation as a weighted average

For each value of Y:

- bars show

$$p(x \mid Y = y)$$

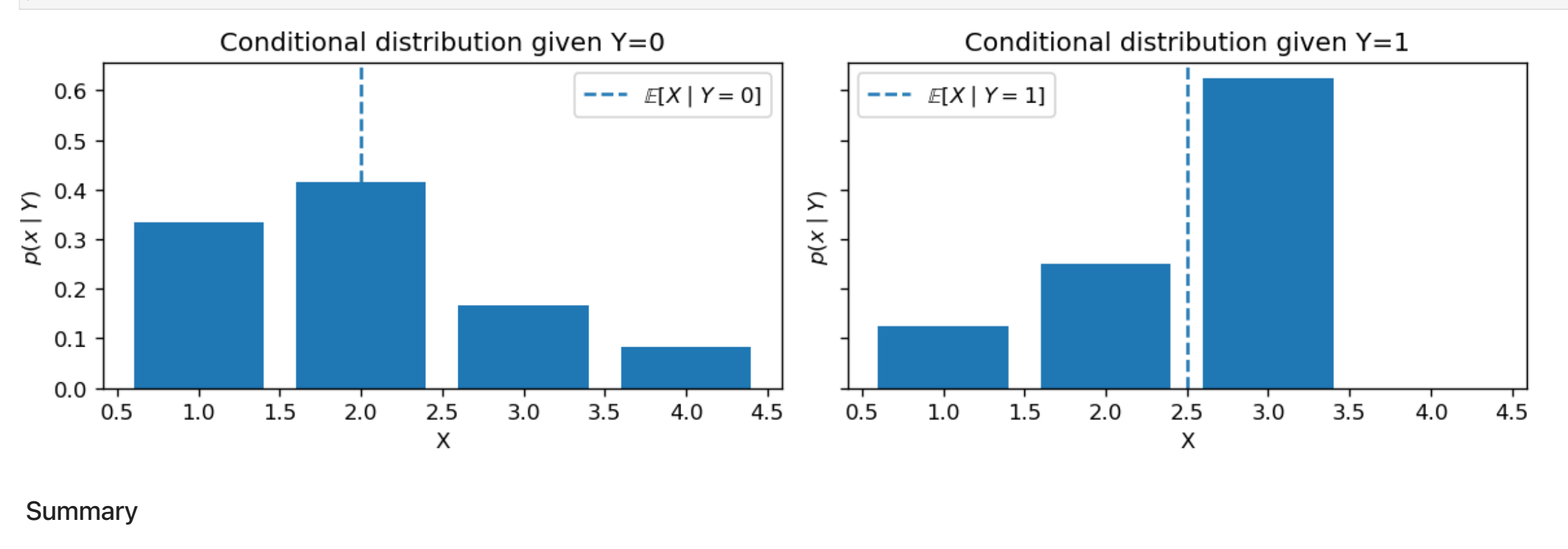
- the dashed line marks

$$\mathbb{E}[X \mid Y = y]$$

This is the **discrete analogue** of “averaging X inside each Y bin.”

In [37]:

```python
fig, axes = plt.subplots(1, 2, figsize=(10, 3), sharey=True)

for j, y in enumerate(Y_vals):
    ax = axes[j]
    ax.bar(X_vals, p_x_given_y[:,j])
    ax.axvline(E_X_given_Y[j], linestyle="--", label=rf"$\mathbb{{E}}[X\mid Y={y}]$")
    ax.set_title(f"Conditional distribution given Y={y}")
    ax.set_xlabel("X")
    ax.set_ylabel(r"$p(x\mid Y)$")
    ax.legend()

plt.tight_layout()
plt.show()
```



Summary

- $\mathbb{E}[X \mid Y]$ is **not a number** but it is a **function of** Y
- In this example: $\mathbb{E}[X \mid Y = 0] \neq \mathbb{E}[X \mid Y = 1]$
- For discrete variables:
  - **Fix** $Y = y$
  - Normalize the column $p(x, y)$ to get $p(x \mid y)$
  - Compute a **weighted average of** X

4. Marginal Expectation (Law of Total Expectation)

$$\mathbb{E}[X] = \mathbb{E}_Y\,[\mathbb{E}[X \mid Y]]$$

In [38]:

```python
# Simple discrete total expectation example
P_Y = np.array([0.6, 0.4])              # P(Y=0), P(Y=1)
E_X_given_Y = np.array([1.0, 3.0])      # E[X|Y=0], E[X|Y=1]
E_X_total = np.sum(P_Y * E_X_given_Y)
E_X_total
```

Out[38]:

```
1.8000000000000003
```

5. Marginal Distribution and Its Relationship to Expectations

A **marginal distribution** describes the probability behavior of a single random variable **by itself**, ignoring all other variables. It is obtained by “averaging out” (summing or integrating over) the other variables in a joint distribution.

Definition

Suppose we have two random variables X and Y with joint distribution $p(x, y)$.

Discrete case

$$p(x) = \sum_y p(x, y)$$

Continuous case

$$p(x) = \int p(x, y)\,dy$$

The result $p(x)$ is the **marginal distribution of** X.

Intuition

Think of a table or surface describing $p(x, y)$.

- The **joint distribution** tells you how (X, Y) behave together.
- The **marginal distribution** of X collapses this information down to **only** X by summing or integrating over all possible values of Y.

In words:

> A marginal distribution describes what X looks like when you do not care about Y.

Relationship to Expectations

An **expectation** is always computed with respect to a distribution. When we compute $\mathbb{E}[X]$, we use the **marginal distribution of** X, not the joint distribution.

Discrete case

$$\mathbb{E}[X] = \sum_x x\,p(x)$$

Continuous case

$$\mathbb{E}[X] = \int x\,p(x)\,dx$$

Even if X is part of a larger joint system, its expectation depends **only** on its marginal.

Expectation via the joint distribution

The same expectation can be written using the joint distribution:

$$\mathbb{E}[X] = \sum_{x,y} x\,p(x, y)$$

This works because summing over y first produces the marginal:

$$\sum_{x,y} x\,p(x, y) = \sum_x x\left(\sum_y p(x, y)\right) = \sum_x x\,p(x)$$

Start with:

$$\mathbb{E}[X] = \sum_{x,y} x\,p(x, y).$$

Now rearrange the sum:

$$\sum_{x,y} x\,p(x, y) = \sum_x \sum_y x\,p(x, y).$$

Because x does **not** depend on y, we can factor it out:

$$= \sum_x x\left(\sum_y p(x, y)\right).$$

The inner sum:

$$\sum_y p(x, y)$$

is **exactly the marginal distribution of** X, denoted $p(x)$.

So we obtain:

$$\sum_x x\,p(x) = \mathbb{E}[X].$$

This shows that:

> **Marginalization is the step that makes expectations well-defined for individual variables.**

Summary

- A **marginal distribution** isolates one variable from a joint system.
- **Expectations of a variable are always taken with respect to its marginal distribution.**
- Joint distributions contain more information, but expectations reduce that information via marginalization.

Marginal Distributions from a 2D Joint Distribution

Let us look at a **simple 2D discrete example** of a joint distribution $p(x, y)$ and shows how to compute and visualize:

- the **marginal distribution** $p(x) = \sum_y p(x, y)$
- the **marginal distribution** $p(y) = \sum_x p(x, y)$

We use a small grid so the summations can be seen clearly.

In [39]:

```python
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.dpi"] = 120
```

Define a small joint distribution $p(x, y)$

We choose discrete supports:

- $X \in \{1, 2, 3, 4\}$
- $Y \in \{0, 1, 2\}$

and define a valid joint distribution $p(x, y)$ whose entries sum to 1.

In [40]:

```python
X_vals = np.array([1, 2, 3, 4])
Y_vals = np.array([0, 1, 2])

# rows = X, cols = Y
p_xy = np.array([
    [0.05, 0.10, 0.05],
    [0.10, 0.20, 0.05],
    [0.05, 0.20, 0.10],
    [0.05, 0.03, 0.02]
], dtype=float)

print("Sum of all probabilities:", p_xy.sum())
```

```
Sum of all probabilities: 1.0
```

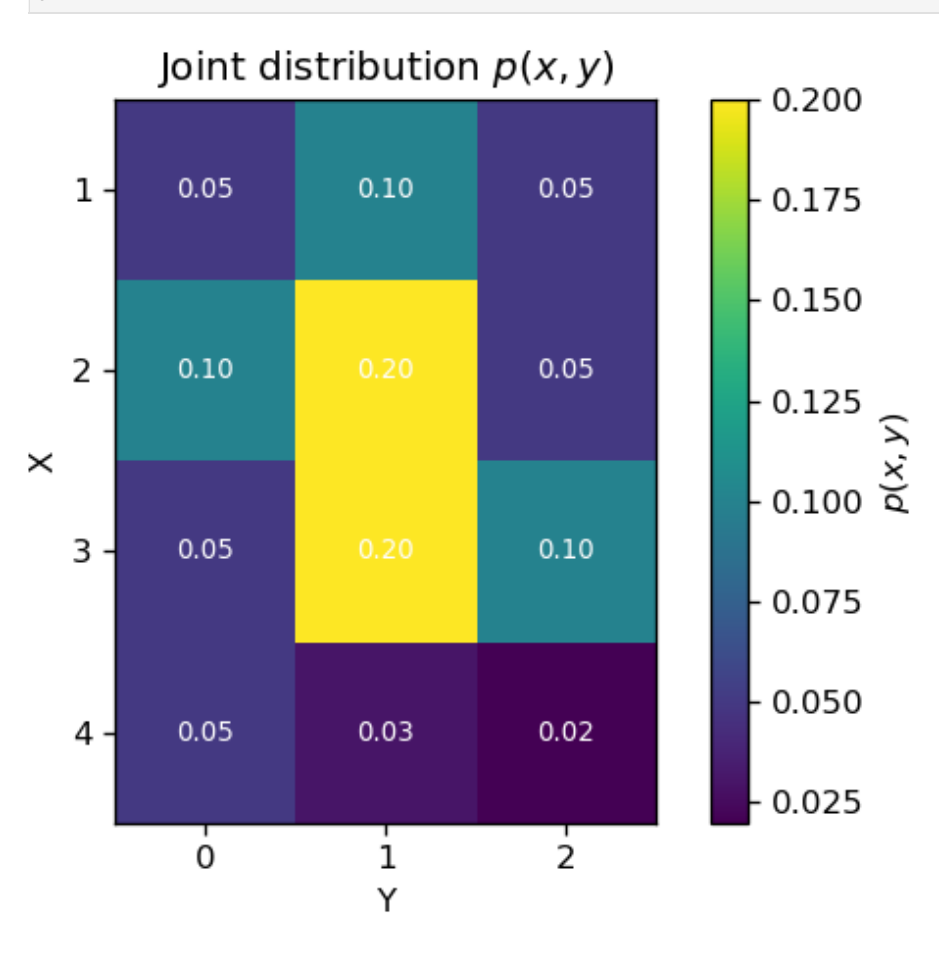
Visualize the joint distribution $p(x, y)$ as a colorful heatmap

Each colored cell is a probability mass $p(x, y)$.

- Brighter color = higher probability mass
- Rows correspond to X values
- Columns correspond to Y values

In [41]:

```python
fig, ax = plt.subplots(figsize=(5.2, 4.0))

im = ax.imshow(p_xy, cmap="viridis")
cbar = fig.colorbar(im, ax=ax)
cbar.set_label("$p(x,y)$")

ax.set_xticks(range(len(Y_vals)))
ax.set_xticklabels(Y_vals)
ax.set_yticks(range(len(X_vals)))
ax.set_yticklabels(X_vals)

ax.set_xlabel("Y")
ax.set_ylabel("X")
ax.set_title("Joint distribution $p(x,y)$")

# Annotate cells with probabilities
for i in range(len(X_vals)):
    for j in range(len(Y_vals)):
        ax.text(j, i, f"{p_xy[i,j]:.2f}", ha="center", va="center", color="white", fontsize=8)

plt.tight_layout()
plt.show()
```



Compute marginals $p(x)$ and $p(y)$

Marginal of (X) (sum over (Y)):

$$p(x) = \sum_y p(x, y)$$

Marginal of (Y) (sum over (X)):

$$p(y) = \sum_x p(x, y)$$

In [42]:

```python
p_x = p_xy.sum(axis=1)  # sum across columns (over Y)
p_y = p_xy.sum(axis=0)  # sum across rows (over X)

print("p(x) =", p_x, " sum =", p_x.sum())
print("p(y) =", p_y, " sum =", p_y.sum())
```

```
p(x) = [0.2  0.35 0.35 0.1 ]  sum = 1.0
p(y) = [0.25 0.53 0.22]  sum = 1.0
```

Show the marginal calculations explicitly (row sums and column sums)

- **Row sums** correspond to $p(x)$
- **Column sums** correspond to $p(y)$

This is exactly what marginalization does: it **collapses** a 2D distribution into 1D.

In [43]:

```python
# Display row and column sums alongside the table in text form
print("Row sums (p(x)):")
for i, xv in enumerate(X_vals):
    print(f"  X={xv}: sum_y p({xv},y) = {p_x[i]:.2f}")

print("\nColumn sums (p(y)):")
for j, yv in enumerate(Y_vals):
    print(f"  Y={yv}: sum_x p(x,{yv}) = {p_y[j]:.2f}")
```

```
Row sums (p(x)):
  X=1: sum_y p(1,y) = 0.20
  X=2: sum_y p(2,y) = 0.35
  X=3: sum_y p(3,y) = 0.35
  X=4: sum_y p(4,y) = 0.10

Column sums (p(y)):
  Y=0: sum_x p(x,0) = 0.25
  Y=1: sum_x p(x,1) = 0.53
  Y=2: sum_x p(x,2) = 0.22
```

Side-by-side visualization: joint + marginals

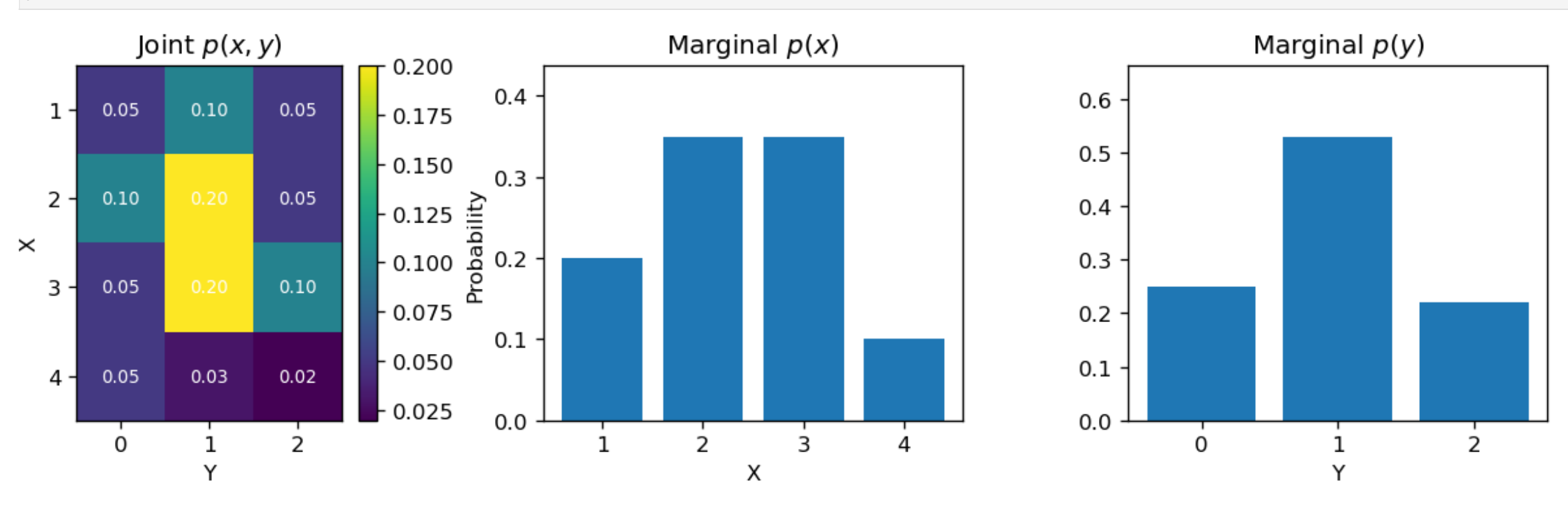
- Left: joint heatmap $p(x, y)$
- Middle: marginal bar chart $p(x)$
- Right: marginal bar chart $p(y)$

This is the cleanest visual interpretation of marginalization:

- sum rows $\to p(x)$
- sum columns $\to p(y)$

In [44]:

```python
fig, axes = plt.subplots(1, 3, figsize=(10.5, 3.2))

# Joint heatmap
ax = axes[0]
im = ax.imshow(p_xy, cmap="viridis")
ax.set_title("Joint $p(x,y)$")
ax.set_xlabel("Y")
ax.set_ylabel("X")
ax.set_xticks(range(len(Y_vals)))
ax.set_xticklabels(Y_vals)
ax.set_yticks(range(len(X_vals)))
ax.set_yticklabels(X_vals)

for i in range(len(X_vals)):
    for j in range(len(Y_vals)):
        ax.text(j, i, f"{p_xy[i,j]:.2f}", ha="center", va="center", color="white", fontsize=8)

fig.colorbar(im, ax=ax, fraction=0.046, pad=0.04)

# Marginal p(x)
ax = axes[1]
ax.bar(X_vals, p_x)
ax.set_title("Marginal $p(x)$")
ax.set_xlabel("X")
ax.set_ylabel("Probability")
ax.set_ylim(0, max(p_x)*1.25)

# Marginal p(y)
ax = axes[2]
ax.bar(Y_vals, p_y)
ax.set_title("Marginal $p(y)$")
ax.set_xlabel("Y")
ax.set_ylim(0, max(p_y)*1.25)

plt.tight_layout()
plt.show()
```



Connect marginals to expectations

Once we have the marginal distributions, expectations follow immediately.

For example:

$$\mathbb{E}[X] = \sum_x x\,p(x), \quad \mathbb{E}[Y] = \sum_y y\,p(y).$$

In [45]:

```python
E_X = np.sum(X_vals * p_x)
E_Y = np.sum(Y_vals * p_y)

print("E[X] =", E_X)
print("E[Y] =", E_Y)
```

```
E[X] = 2.35
E[Y] = 0.97
```

Summary

- A **joint** distribution $p(x, y)$ describes how (X, Y) behave together.
- The **marginal** distributions $p(x)$ and $p(y)$ describe each variable alone.
- Marginals are computed by **summing out** the other variable.

6 Conditionals and Bayes' rule

From the joint distribution:

$$p(x, y) = p(x \mid y)p(y) = p(y \mid x)p(x)$$

Rearranging gives **Bayes' rule**:

$$p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)}$$

Bayes' rule simply expresses **two ways of slicing the same joint distribution**.

In [46]:

```python
# Conditional distributions
p_x_given_y = p_xy / p_y          # p(x|y)
p_y_given_x = (p_xy.T / p_x).T    # p(y|x)

print("p(x|y=0):", p_x_given_y[:,0])
print("p(x|y=1):", p_x_given_y[:,1])
print("p(y|x=0):", p_y_given_x[0])
print("p(y|x=1):", p_y_given_x[1])
```

```
p(x|y=0): [0.2 0.4 0.2 0.2]
p(x|y=1): [0.18867925 0.37735849 0.37735849 0.05660377]
p(y|x=0): [0.25 0.5  0.25]
p(y|x=1): [0.28571429 0.57142857 0.14285714]
```

Expectations use marginal distributions

The expectation of a random variable depends **only on its marginal distribution**.

$$\mathbb{E}[X] = \sum_x x\,p(x)$$

In [51]:

```python
E_X = np.sum(X_vals * p_x)
E_Y = np.sum(Y_vals * p_y)

print("E[X] =", E_X)
print("E[Y] =", E_Y)
```

```
E[X] = 2.35
E[Y] = 0.97
```

Conditional expectations

The conditional expectation of X given $Y = y$ is:

$$\mathbb{E}[X\mid Y=y]=\sum_x x\,p(x\mid y)$$

This is a **different number for each value of (Y)**.

In [49]:

```
E_X_given_y0 = np.sum(X_vals * p_x_given_y[:,0])
E_X_given_y1 = np.sum(X_vals * p_x_given_y[:,1])

print("E[X | Y=0] =", E_X_given_y0)
print("E[X | Y=1] =", E_X_given_y1)
```

```
E[X | Y=0] = 2.4000000000000004
E[X | Y=1] = 2.30188679245283
```

Law of Total Expectation

The **law of total expectation** connects everything:

$$\mathbb{E}[X]=\sum_y \mathbb{E}[X\mid Y=y]\,p(y)$$

This says:

> The overall average of (X) is a **weighted average of its conditional averages.**

In [52]:

```
E_X_total = E_X_given_y0 * p_y[0] + E_X_given_y1 * p_y[1]

print("E[X] from total expectation =", E_X_total)
```

```
E[X] from total expectation = 1.82
```

Summary

- **Joint distribution**: complete description of (X, Y)
- **Marginals**: collapse the joint by summing
- **Conditionals**: slice and renormalize the joint
- **Bayes' rule**: relates conditionals via marginals
- **Law of total expectation**: averages conditional expectations

All of these are different views of the **same probability object**.

Law of Total Expectation Example - Exam Scores by Section

Assume:

- Y = section (Morning vs Evening)
- X = exam score

The law states:

$$\mathbb{E}[X]=\sum_y \mathbb{E}[X\mid Y=y]\,p(y).$$

In words:

> The overall average score equals the weighted average of the section averages.

We visualize:

1. Section sizes $p(Y)$
2. Section averages $\mathbb{E}[X\mid Y]$
3. The weighted contributions $p(Y)\,\mathbb{E}[X\mid Y]$
4. The overall mean $\mathbb{E}[X]$
5. A small synthetic dataset to verify the same idea by direct averaging

In [53]:

```
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.dpi"] = 120
np.random.seed(0)
```

Setup: Section proportions and section average scores

We assume:

- $p(Y=\text{Morning})=0.6$
- $p(Y=\text{Evening})=0.4$

And conditional (within-section) average scores:

- $\mathbb{E}[X\mid Y=\text{Morning}]=80$
- $\mathbb{E}[X\mid Y=\text{Evening}]=70$

In [54]:

```
sections = ["Morning", "Evening"]
pY = np.array([0.6, 0.4])          # p(Y)
EX_given_Y = np.array([80, 70])    # E[X | Y]

EX_total = np.sum(pY * EX_given_Y)

print("E[X] by total expectation =", EX_total)
```

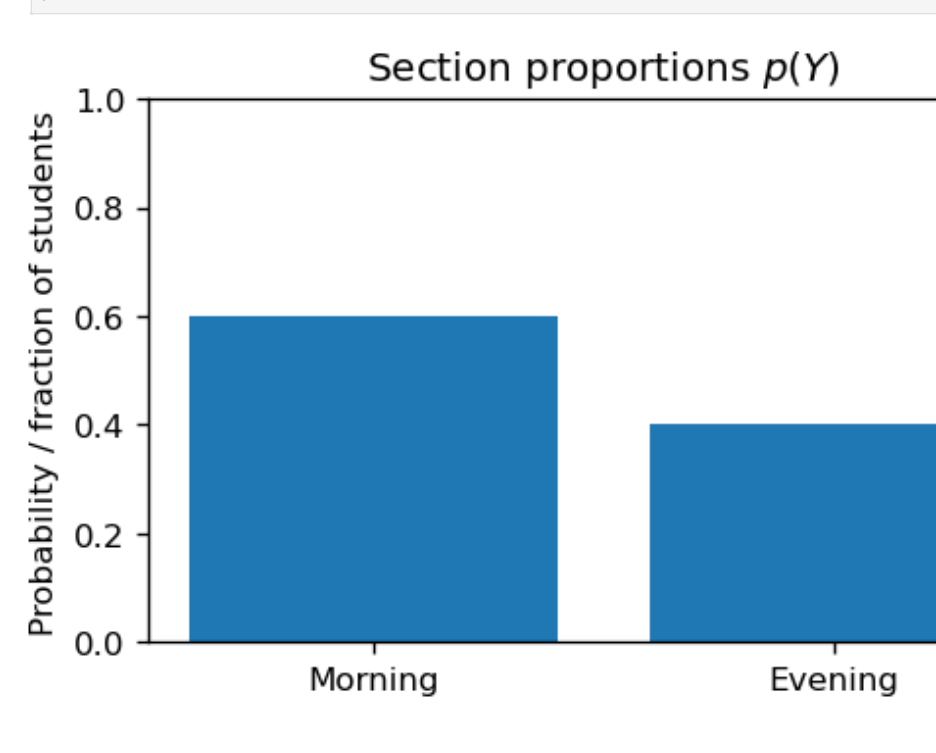
```
E[X] by total expectation = 76.0
```

Visual 1: Section proportions $p(Y)$

This plot shows how the population is split between sections.

In [55]:

```
plt.figure(figsize=(5,3))
plt.bar(sections, pY)
plt.ylim(0, 1)
plt.ylabel("Probability / fraction of students")
plt.title("Section proportions $p(Y)$")
plt.show()
```

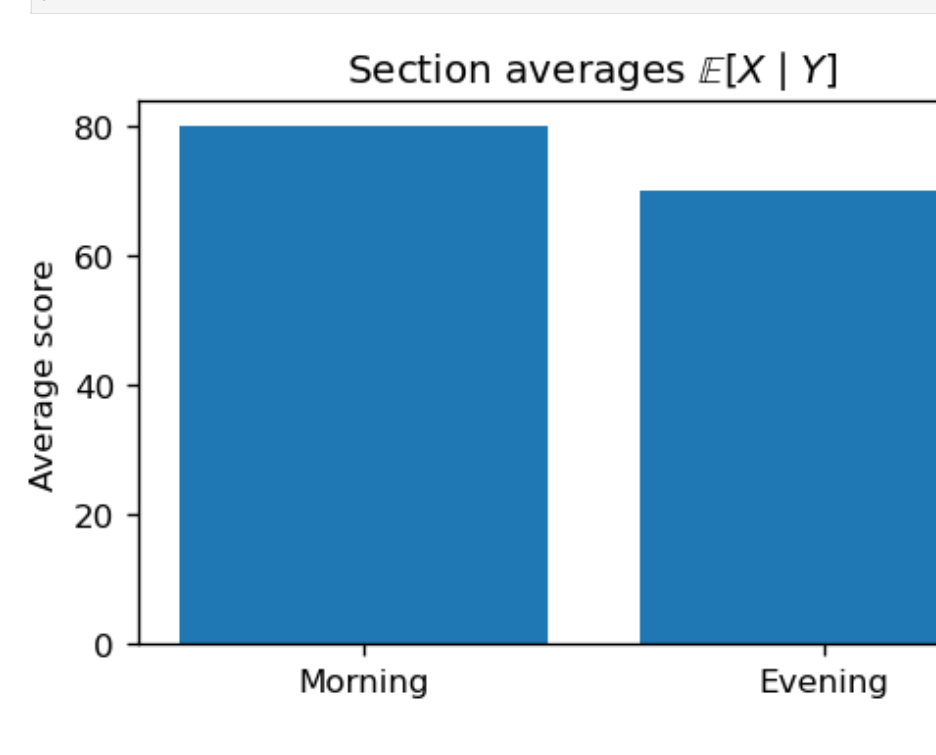


Visual 2: Section averages $\mathbb{E}[X\mid Y]$

This plot shows the average score **within each section**.

In [56]:

```
plt.figure(figsize=(5,3))
plt.bar(sections, EX_given_Y)
plt.ylabel("Average score")
plt.title("Section averages $\mathbb{E}[X\mid Y]$")
plt.show()
```



Visual 3: Weighted-average picture (the law of total expectation)

We compute:

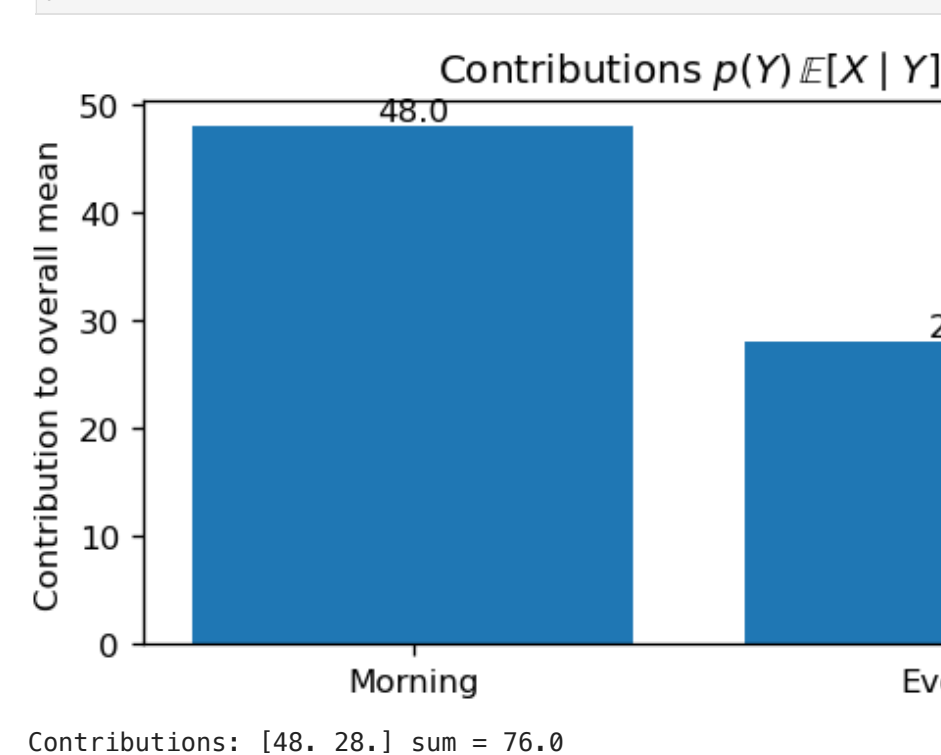
$$\mathbb{E}[X]=0.6\cdot 80+0.4\cdot 70=76.$$

Each section contributes:

- Morning contributes $0.6\times 80=48$
- Evening contributes $0.4\times 70=28$

In [57]:

```
contrib = pY * EX_given_Y

plt.figure(figsize=(6,3))
plt.bar(sections, contrib)
plt.ylabel("Contribution to overall mean")
plt.title("Contributions $p(Y)\,\mathbb{E}[X\mid Y]$")

for i, v in enumerate(contrib):
    plt.text(i, v + 0.6, f"{v:.1f}", ha="center")

plt.show()

print("Contributions:", contrib, "sum =", contrib.sum())
```



```
Contributions: [48. 28.] sum = 76.0
```

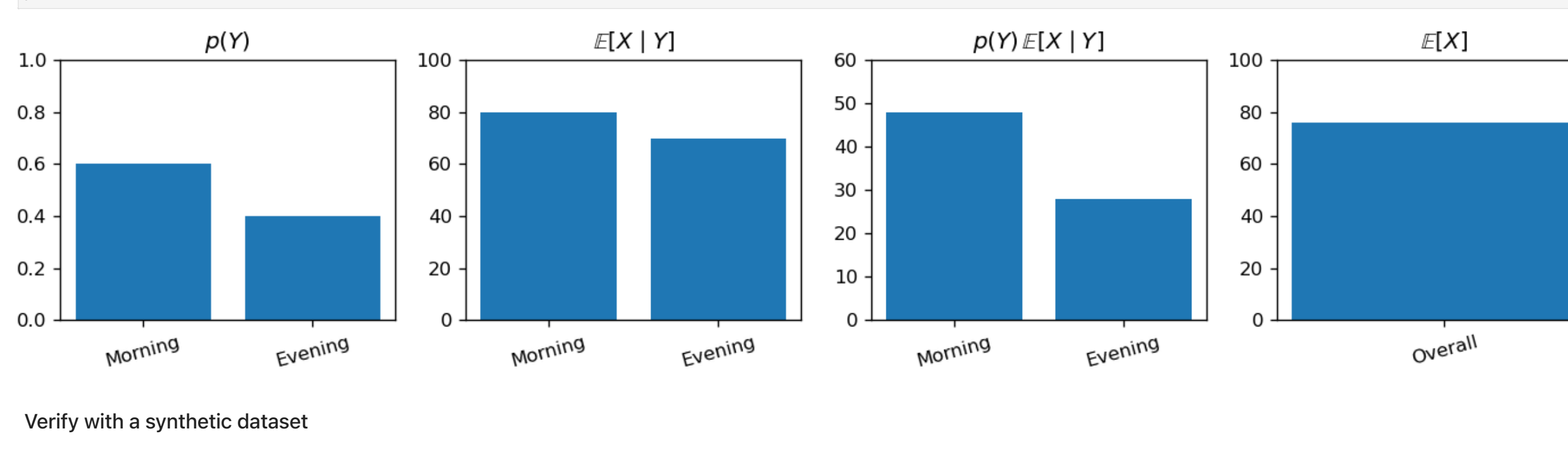
Visual 4: Put it all together

Side-by-side:

- $p(Y)$
- $\mathbb{E}[X\mid Y]$
- $p(Y)\,\mathbb{E}[X\mid Y]$
- $\mathbb{E}[X]$

In [58]:

```
fig, axes = plt.subplots(1, 4, figsize=(12, 2.8))

axes[0].bar(sections, pY)
axes[0].set_ylim(0, 1)
axes[0].set_title("$p(Y)$")

axes[1].bar(sections, EX_given_Y)
axes[1].set_title("$\mathbb{E}[X\mid Y]$")
axes[1].set_ylim(0, 100)

axes[2].bar(sections, contrib)
axes[2].set_title("$p(Y)\,\mathbb{E}[X\mid Y]$")
axes[2].set_ylim(0, 60)

axes[3].bar(["Overall"], [EX_total])
axes[3].set_title("$\mathbb{E}[X]$")
axes[3].set_ylim(0, 100)

for ax in axes:
    ax.tick_params(axis='x', rotation=15)

plt.tight_layout()
plt.show()
```



Verify with a synthetic dataset

To make the law feel concrete, we can simulate students:

- 60% are assigned to Morning, scores centered near 80
- 40% are assigned to Evening, scores centered near 70

Then we compare:

- the **direct average** of all scores
- the **weighted average** of section means

These match closely (up to sampling noise).

In [59]:

```
N = 400
Y = np.random.choice([0, 1], size=N, p=pY)  # 0=Morning, 1=Evening

# Generate scores with some spread
scores = np.where(Y==0,
                  np.random.normal(loc=80, scale=8, size=N),
                  np.random.normal(loc=70, scale=8, size=N))

# Direct overall average
EX_direct = scores.mean()

# Conditional means from the simulated data
EX_morning = scores[Y==0].mean()
EX_evening = scores[Y==1].mean()

# Weighted average using empirical p(Y)
pY_emp = np.array([(Y==0).mean(), (Y==1).mean()])
EX_total_emp = pY_emp[0]*EX_morning + pY_emp[1]*EX_evening

print("Direct average of all scores:", EX_direct)
print("Empirical E[X|Morning], E[X|Evening]:", EX_morning, EX_evening)
print("Empirical p(Y):", pY_emp)
print("Weighted average (total expectation):", EX_total_emp)
```

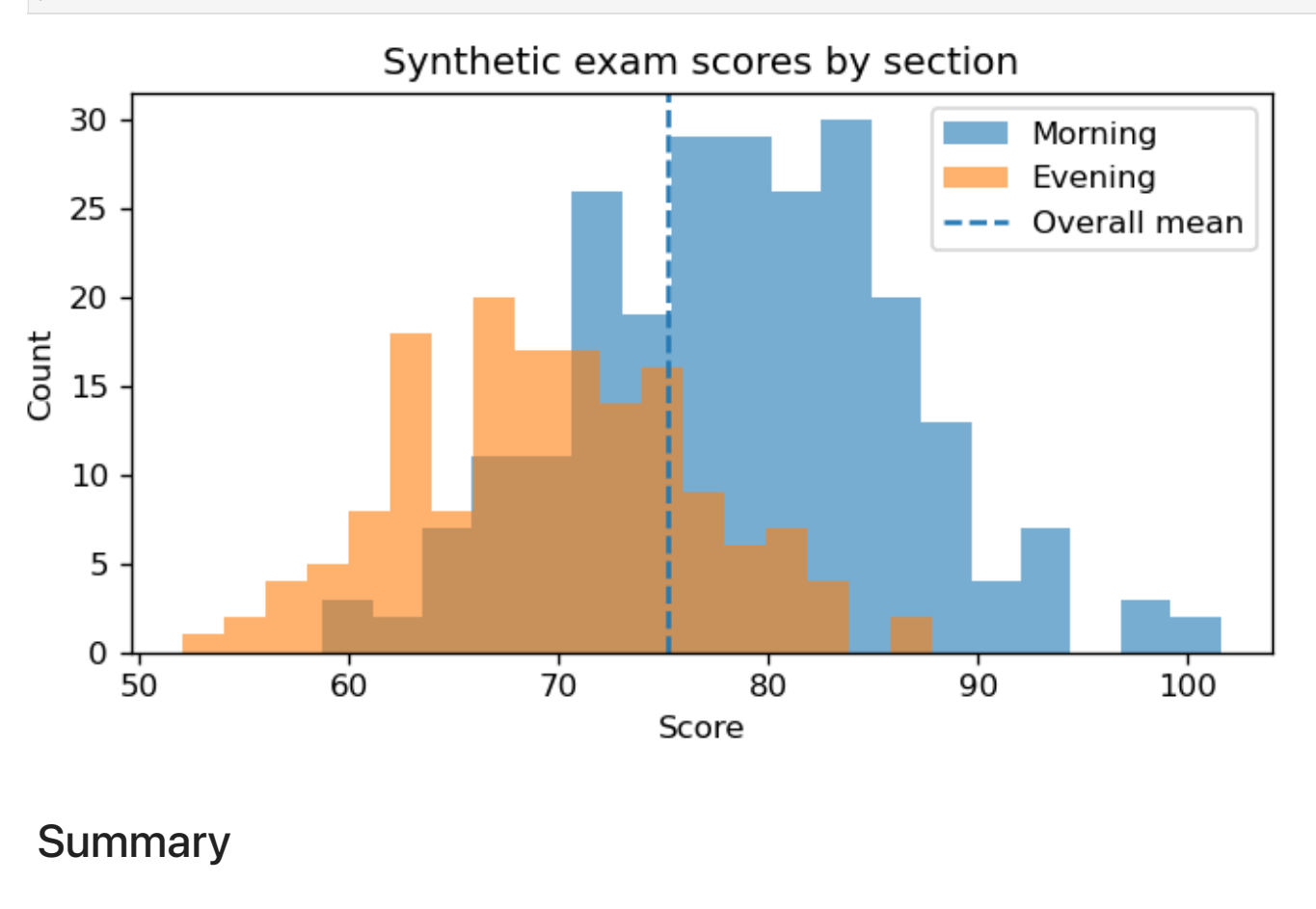
```
Direct average of all scores: 75.20547083584216
Empirical E[X|Morning], E[X|Evening]: 78.80764626289046 69.68821480200876
Empirical p(Y): [0.605 0.395]
Weighted average (total expectation): 75.20547083584219
```

Visual 5: Synthetic score distributions by section

We plot histograms of scores for each section and mark the overall mean.

In [24]:

```
plt.figure(figsize=(6.5,3.2))
plt.hist(scores[Y==0], bins=18, alpha=0.6, label="Morning")
plt.hist(scores[Y==1], bins=18, alpha=0.6, label="Evening")
plt.axvline(scores.mean(), linestyle="--", label="Overall mean")

plt.xlabel("Score")
plt.ylabel("Count")
plt.title("Synthetic exam scores by section")
plt.legend()
plt.show()
```



Summary

The law of total expectation says:

$$\mathbb{E}[X]=\sum_y \mathbb{E}[X\mid Y=y]\,p(y).$$

Visually:

- the **section proportions** provide the weights $p(y)$
- the **section averages** provide the conditional means $\mathbb{E}[X\mid Y=y]$
- the overall mean is a **weighted average of group means**

5. Variance, Covariance, Correlation

Variance

$$\mathrm{Var}(X)=\mathbb{E}\big[(X-\mathbb{E}[X])^2\big]=\mathbb{E}[X^2]-(\mathbb{E}[X])^2$$

Covariance

$$\mathrm{Cov}(X,Y)=\mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]$$

Correlation

$$\rho_{XY}=\frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

In [ ]:

```
# E[X^2] vs (E[X])^2
Z = np.random.normal(size=200_000)
EX2 = np.mean(Z**2)
EX  = np.mean(Z)
(EX2, EX**2, EX2-EX**2)
```

Out[ ]:

```
(0.9962321669692806, 1.1695788970707868e-05, 0.9962204711803099)
```

In [ ]:

```
# Covariance/correlation example
X = np.random.normal(size=200_000)
Y = 2*X + np.random.normal(scale=0.5, size=200_000)

cov = np.mean((X-np.mean(X))*(Y-np.mean(Y)))
corr = cov / (np.std(X)*np.std(Y))
cov, corr
```

Out[ ]:

```
(1.9974464048528084, 0.9700904439349247)
```

6. Law of Total Variance

$$\mathrm{Var}(X)=\mathbb{E}[\mathrm{Var}(X\mid Y)]+\mathrm{Var}(\mathbb{E}[X\mid Y])$$

What the next Monte Carlo example illustrates

This simulation provides a **numerical verification** of the **Law of Total Variance**:

How the example is constructed

- A binary variable $Y\in\{0,1\}$ is sampled with:

$$P(Y=1)=0.4,\quad P(Y=0)=0.6$$

- The variable X is generated **conditionally on** Y:
  - If $Y=0$, then $X\sim\mathcal{N}(0,1)$
  - If $Y=1$, then $X\sim\mathcal{N}(3,2^2)$

This creates two groups with **different means and different variances**.

What each computed quantity represents

- `var_total`
  The empirical variance of **all samples of** X, ignoring Y.
- `var_within`
  The **average of the conditional variances**: $\mathbb{E}[\mathrm{Var}(X\mid Y)]$ This captures the variability **within each group** where Y defines the group. In other words, what is the average variability across groups defined by Y.
- `var_between`
  The variance of the conditional means: $\mathrm{Var}(\mathbb{E}[X\mid Y])$ This captures the variability **between groups** caused by different group means. In other words, what is the average variability across the means of each group.

Summary

The output shows that: $\texttt{var\_total}\approx\texttt{var\_within}+\texttt{var\_between}$

This confirms that the total variability in X can be decomposed into:

- **within-group randomness**, and
- **between-group differences in the conditional mean**.

In [62]:

```
# Monte Carlo illustration of total variance
N = 300_000
Y = np.random.binomial(1, 0.4, size=N)  # group indicator

# X|Y=0 ~ N(0,1), X|Y=1 ~ N(3,2)
X = np.where(Y==0, np.random.normal(0,1,size=N), np.random.normal(3,2,size=N))

var_total = np.var(X)
var_within = 0.5*(np.var(X[Y==0]) + np.var(X[Y==1]))
mean0, mean1 = np.mean(X[Y==0]), np.mean(X[Y==1])
var_between = np.var(np.where(Y==0, mean0, mean1))

var_total, var_within + var_between
```

Out[62]:

```
(4.355836629866026, 4.650022890963639)
```

Convex and Concave Functions (Geometric Intuition)

A function's convexity or concavity describes **how it bends**.

Convex function

A function $\phi(x)$ is **convex** if the line segment between any two points on the graph lies **above** the function.

Formally, for all x_1, x_2 and $\lambda\in[0,1]$:

$$\phi(\lambda x_1+(1-\lambda)x_2)\ \le\ \lambda\phi(x_1)+(1-\lambda)\phi(x_2)$$

Visual intuition

- Draw two points on the curve
- Draw the straight line (chord) between them
- The curve stays **below** the line

Common convex examples

- $\phi(x)=x^2$
- $\phi(x)=|x|$
- $\phi(x)=e^x$

Concave function

A function $\phi(x)$ is **concave** if the line segment between any two points on the graph lies **below** the function.

Formally:

$$\phi(\lambda x_1+(1-\lambda)x_2)\ \ge\ \lambda\phi(x_1)+(1-\lambda)\phi(x_2)$$

Visual intuition

- Draw two points on the curve
- Draw the straight line between them
- The curve stays **above** the line

Common concave examples

- $\phi(x)=\log x$ (for $x>0$)
- $\phi(x)=\sqrt{x}$
- $\phi(x)=-x^2$

7. Jensen's Inequality

If ϕ is **convex**:

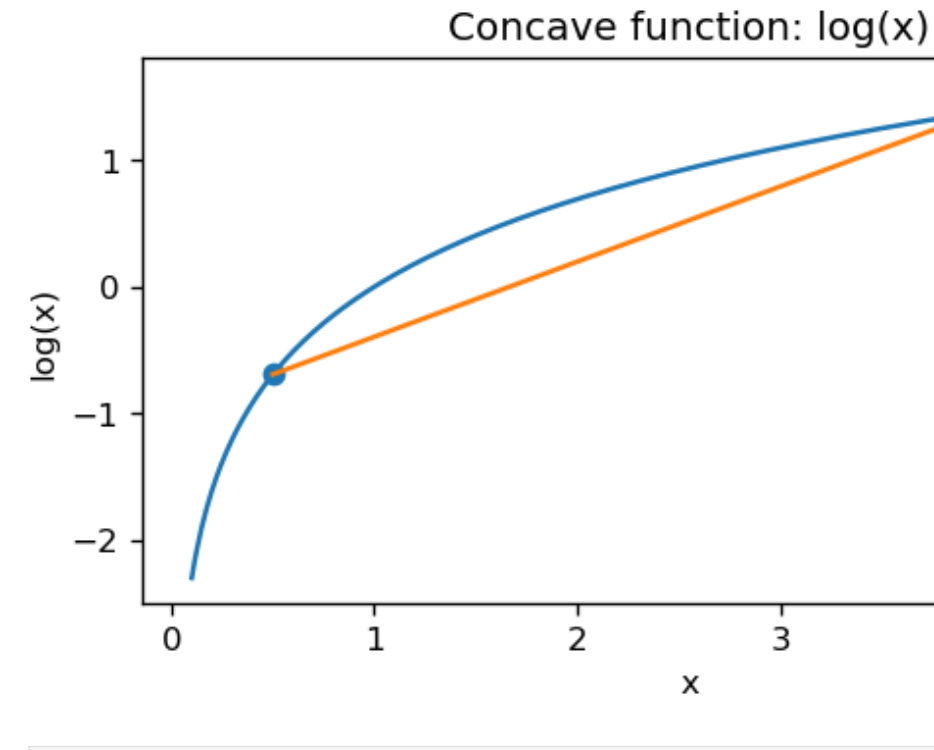
$$\phi(\mathbb{E}[X])\le\mathbb{E}[\phi(X)]$$

If ϕ is **concave** (e.g., $\log$), the inequality flips:

$$\mathbb{E}[\log X]\le\log\mathbb{E}[X]$$

In [63]:

```
# Visual Jensen for log
xs = np.linspace(0.1, 5, 400)
fig, ax = plt.subplots(figsize=(6,3))
ax.plot(xs, np.log(xs))
ax.set_title("Concave function: log(x)")
ax.set_xlabel("x")
ax.set_ylabel("log(x)")

a, b = 0.5, 4.0
ax.plot([a,b], [np.log(a), np.log(b)])  # chord
ax.scatter([a,b], [np.log(a), np.log(b)])
plt.show()
```



In [64]:

```
# Jensen demo with log (concave)
X = np.random.lognormal(mean=0.0, sigma=1.0, size=200_000)  # positive
lhs = np.mean(np.log(X))
rhs = np.log(np.mean(X))
lhs, rhs, lhs <= rhs
```

Out[64]:

```
(0.000632538165453497, 0.49687765422158064, True)
```

8. Entropy and Cross-Entropy

Entropy

$$H(P)=-\sum_x p(x)\log p(x)$$

Cross-entropy

$$H(P,Q)=-\sum_x p(x)\log q(x)$$

In [65]:

```
p = np.array([0.1, 0.9])
q = np.array([0.5, 0.5])

H_p = -np.sum(p*np.log(p))
H_pq = -np.sum(p*np.log(q))
H_p, H_pq
```

Out[65]:

```
(0.3250829733914482, 0.6931471805599453)
```

Why entropy has the form $-\sum_x p(x)\log p(x)$

Entropy measures **uncertainty** or **surprise** in a probability distribution.

Step 1 — Why use $\log$?

We want a measure of surprise for a single outcome x that:

- is **small** when $p(x)$ is large (likely outcomes are not surprising),
- is **large** when $p(x)$ is small (rare outcomes are surprising),
- adds nicely across independent events.

The function $-\log p(x)$ satisfies all three:

- $p(x)\approx 1\Rightarrow -\log p(x)\approx 0$
- $p(x)\ll 1\Rightarrow -\log p(x)\gg 0$

This quantity is called the **information content** (or surprise) of outcome x.

Step 2 — Why multiply by $p(x)$?

Entropy is the **average surprise**: $H(P)=\mathbb{E}[-\log p(X)]$ Writing this expectation explicitly for a discrete distribution gives: $H(P)=-\sum_x p(x)\log p(x)$

So:

- $\log$ measures **surprise**
- $p(x)$ weights how often that surprise occurs

Entropy is not about the worst case but about the **typical** case.

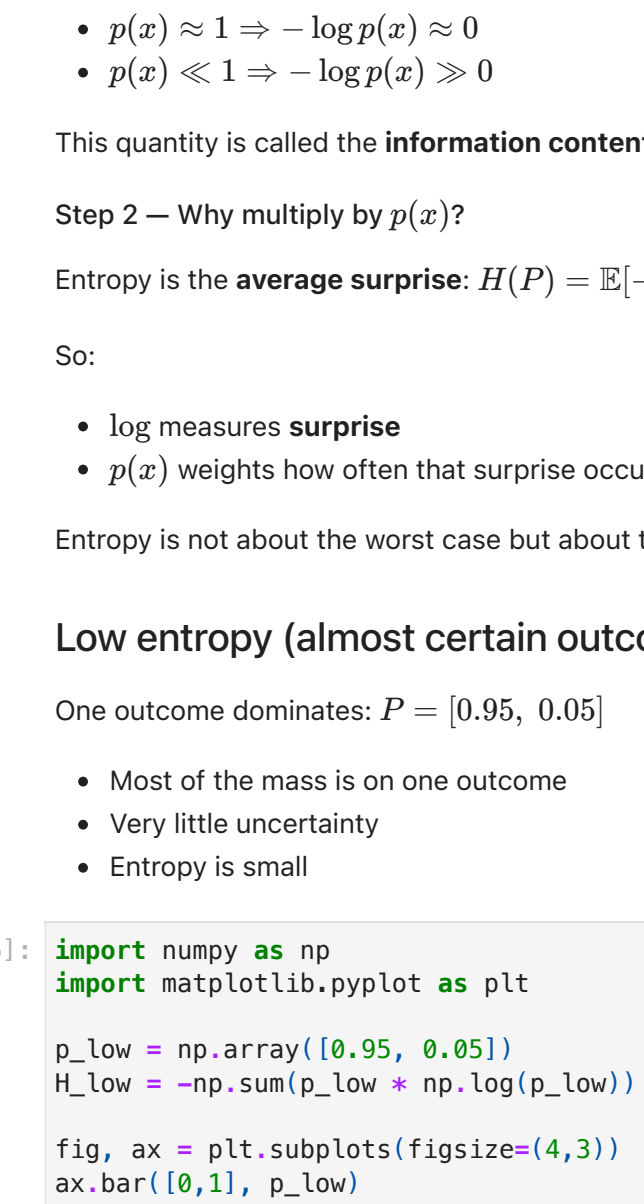
Low entropy (almost certain outcome)

One outcome dominates: $P=[0.95,\ 0.05]$

- Most of the mass is on one outcome
- Very little uncertainty
- Entropy is small

In [66]:

```
import numpy as np
import matplotlib.pyplot as plt

p_low = np.array([0.95, 0.05])
H_low = -np.sum(p_low * np.log(p_low))

fig, ax = plt.subplots(figsize=(4,3))
ax.bar([0,1], p_low)
ax.set_ylim(0,1)
ax.set_xticks([0,1])
ax.set_xticklabels(["A","B"])
ax.set_title(f"Low entropy distribution\nH ≈ {H_low:.3f}")
ax.set_ylabel("p(x)")
plt.show()
```



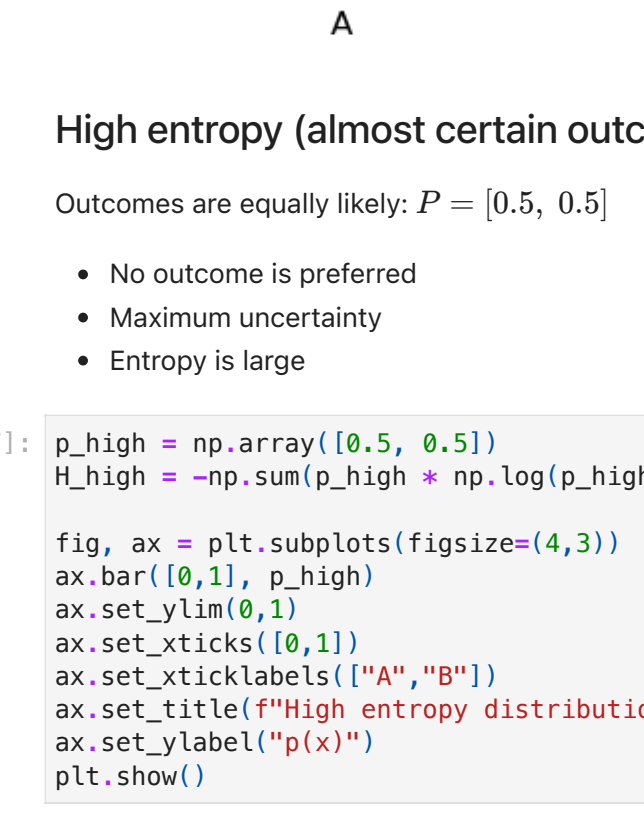
High entropy (almost certain outcome)

Outcomes are equally likely: $P=[0.5,\ 0.5]$

- No outcome is preferred
- Maximum uncertainty
- Entropy is large

In [67]:

```
p_high = np.array([0.5, 0.5])
H_high = -np.sum(p_high * np.log(p_high))

fig, ax = plt.subplots(figsize=(4,3))
ax.bar([0,1], p_high)
ax.set_ylim(0,1)
ax.set_xticks([0,1])
ax.set_xticklabels(["A","B"])
ax.set_title(f"High entropy distribution\nH ≈ {H_high:.3f}")
ax.set_ylabel("p(x)")
plt.show()
```



9. Kullback–Leibler (KL) Divergence

$\mathrm{KL}(P\|Q)=\sum_x p(x)\log\frac{p(x)}{q(x)}$

Identity

$\mathrm{KL}(P\|Q)=H(P,Q)-H(P)$

In [68]:

```
p = np.array([0.4, 0.6])
q = np.array([0.5, 0.5])

kl_pq = np.sum(p*np.log(p/q))
kl_qp = np.sum(q*np.log(q/p))
kl_pq, kl_qp
```

Out[68]:

```
(0.020135513550688877, 0.020410997260127586)
```

Kullback–Leibler (KL) Divergence and Its Relation to Cross-Entropy

KL divergence measures **how different** one probability distribution is from another.

Definition

For two discrete distributions P (true) and Q (model):

$$\mathrm{KL}(P\|Q)=\sum_x p(x)\log\frac{p(x)}{q(x)}$$

Equivalently, as an expectation under P:

$$\mathrm{KL}(P\|Q)=\mathbb{E}_{x\sim P}\left[\log\frac{p(x)}{q(x)}\right]$$

Interpretation

- P is the distribution that **actually generates the data**
- Q is the distribution we **use to model or encode** that data

KL divergence answers the question:

> *How much extra information (on average) do we need when we use Q instead of the true distribution P?*

Why KL is always non-negative

KL divergence satisfies: $\mathrm{KL}(P\|Q)\ge 0$ with equality if and only if: $P=Q$

This means:

- Using the true distribution is optimal
- Any mismatch introduces an **information penalty**

Connection to Cross-Entropy

Cross-Entropy Definition

The cross-entropy between P and Q is: $H(P,Q)=-\sum_x p(x)\log q(x)$

This is the **expected code length** when data comes from P but we encode it using Q.

Key Identity (Very Important)

KL divergence decomposes cross-entropy as:

$$\mathrm{KL}(P\|Q)=H(P,Q)-H(P)$$

where:

- $H(P)$ is the entropy of the data (irreducible uncertainty)
- $H(P,Q)$ is the total uncertainty when using model Q

Intuitive takeaway

- **Entropy** $H(P)$
  → uncertainty inherent in the data
- **Cross-entropy** $H(P,Q)$
  → uncertainty + model mismatch
- **KL divergence**
  → *pure penalty* for using the wrong model

Why "pure penalty"?

Notice that:

- $H(P)$ depends **only on the data**
- It does **not depend on the model** Q

So when we subtract $H(P)$ from $H(P,Q)$, everything left is due to **model mismatch**.

That is:

$$\text{penalty from using } Q \text{ instead of } P = H(P,Q)-H(P)$$

This penalty is **pure** because:

- it is zero when $Q=P$,
- it increases as Q diverges from P,
- it does not mix in data uncertainty.

So:

> Minimizing cross-entropy is equivalent to minimizing KL divergence from the data distribution to the model.

Why this matters in machine learning

Why this matters in machine learning

In practice:

- P is the empirical data distribution (often one-hot labels)
- Q is the model output (softmax probabilities)

Training with cross-entropy loss is therefore equivalent to: $\min_Q \mathrm{KL}(P\|Q)$

Final intuition

> **Entropy is the unavoidable cost.**
> **KL divergence is the avoidable cost.**

Everything learning can improve lives inside the KL term.

Visual example: KL divergence for close vs. far distributions

We compare two cases using **three-bin discrete distributions**:

- **Close distributions** → small KL divergence
- **Far distributions** → large KL divergence

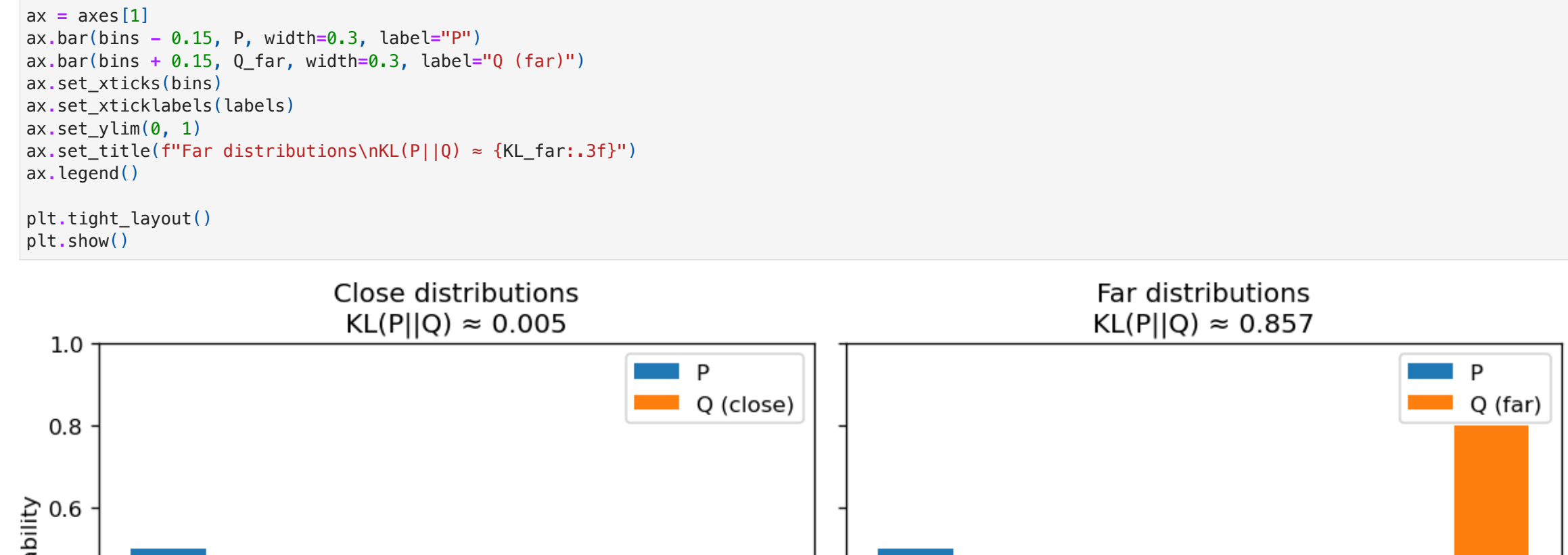
Recall:

$$\mathrm{KL}(P\|Q) = \sum_x p(x)\log\frac{p(x)}{q(x)}$$

KL is an **expectation under** P, so bins where P is large matter most.

In [69]:

```python
import numpy as np
import matplotlib.pyplot as plt

# Three-bin distributions
P = np.array([0.5, 0.3, 0.2])

Q_close = np.array([0.45, 0.33, 0.22])   # close to P
Q_far   = np.array([0.1, 0.1, 0.8])      # very different from P

def kl(P, Q):
    return np.sum(P * np.log(P / Q))

KL_close = kl(P, Q_close)
KL_far   = kl(P, Q_far)

bins = np.arange(3)
labels = ["Bin 1", "Bin 2", "Bin 3"]

fig, axes = plt.subplots(1, 2, figsize=(10, 3.5), sharey=True)

# ---- Close distributions ----
ax = axes[0]
ax.bar(bins - 0.15, P, width=0.3, label="P")
ax.bar(bins + 0.15, Q_close, width=0.3, label="Q (close)")
ax.set_xticks(bins)
ax.set_xticklabels(labels)
ax.set_ylim(0, 1)
ax.set_title(f"Close distributions\nKL(P||Q) ≈ {KL_close:.3f}")
ax.set_ylabel("Probability")
ax.legend()

# ---- Far distributions ----
ax = axes[1]
ax.bar(bins - 0.15, P, width=0.3, label="P")
ax.bar(bins + 0.15, Q_far, width=0.3, label="Q (far)")
ax.set_xticks(bins)
ax.set_xticklabels(labels)
ax.set_ylim(0, 1)
ax.set_title(f"Far distributions\nKL(P||Q) ≈ {KL_far:.3f}")
ax.legend()

plt.tight_layout()
plt.show()
```



Example: KL between 1D Gaussians

For $P = \mathcal{N}(\mu_p, \sigma_p^2)$ and $Q = \mathcal{N}(\mu_q, \sigma_q^2)$:

$$\mathrm{KL}(P\|Q) = \log\frac{\sigma_q}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_q)^2}{2\sigma_q^2} - \frac{1}{2}$$

In [70]:

```python
def kl_gaussian_1d(mu_p, sig_p, mu_q, sig_q):
    return np.log(sig_q/sig_p) + (sig_p**2 + (mu_p-mu_q)**2)/(2*sig_q**2) - 0.5

kl_gaussian_1d(0.0, 1.0, 1.0, 2.0)
```

Out[70]:

```
0.4431471805599453
```

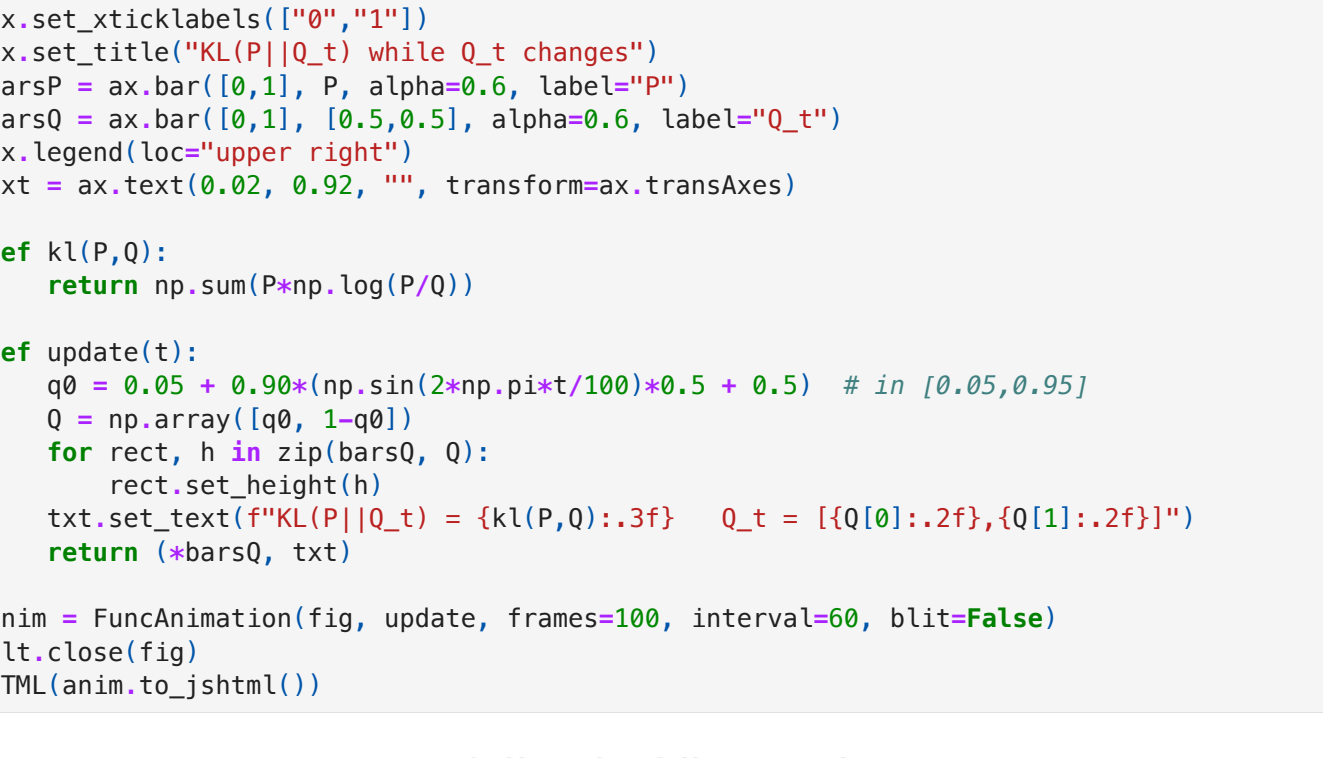
Animation — KL Sensitivity (Discrete)

Fix P and morph Q_t. Watch $\mathrm{KL}(P\|Q_t)$ change.

In [71]:

```python
from matplotlib.animation import FuncAnimation
from IPython.display import HTML

P = np.array([0.6, 0.4])

fig, ax = plt.subplots(figsize=(6,3))
ax.set_ylim(0, 1.0)
ax.set_xticks([0,1])
ax.set_xticklabels(["0","1"])
ax.set_title("KL(P||Q_t) while Q_t changes")
barsP = ax.bar([0,1], P, alpha=0.6, label="P")
barsQ = ax.bar([0,1], [0.5,0.5], alpha=0.6, label="Q_t")
ax.legend(loc="upper right")
txt = ax.text(0.02, 0.92, "", transform=ax.transAxes)

def kl(P,Q):
    return np.sum(P*np.log(P/Q))

def update(t):
    q0 = 0.05 + 0.90*(np.sin(2*np.pi*t/100)*0.5 + 0.5)  # in [0.05,0.95]
    Q = np.array([q0, 1-q0])
    for rect, h in zip(barsQ, Q):
        rect.set_height(h)
    txt.set_text(f"KL(P||Q_t) = {kl(P,Q):.3f}   Q_t = [{Q[0]:.2f},{Q[1]:.2f}]")
    return (*barsQ, txt)

anim = FuncAnimation(fig, update, frames=100, interval=60, blit=False)
plt.close(fig)
HTML(anim.to_jshtml())
```

Out[71]:



Why do we use cross-entropy if it includes the data's natural entropy?

At first glance, cross-entropy seems "wasteful" because it includes the **natural entropy of the data**, which we cannot reduce.

Rewriting KL Divergence Equation: $H(P,Q) = H(P) + \mathrm{KL}(P\|Q)$

So why optimize something that contains an **irreducible term** $H(P)$?

Key insight

> **Because $H(P)$ is constant with respect to the model.**

When training a model, we can only change Q, not the data distribution P.

Minimizing cross-entropy: $\min_Q H(P,Q)$ is therefore exactly equivalent to minimizing: $\min_Q \mathrm{KL}(P\|Q)$. The natural entropy $H(P)$ does not affect optimization because is just a constant offset.

Why we *cannot* directly minimize KL in practice

Although KL divergence is the "clean" objective conceptually, it requires knowledge of: $p(x)$

But:

- the true data distribution P is unknown,
- we only have **samples** from P.

Cross-entropy can be written as an **expectation over data samples**: $H(P,Q) = \mathbb{E}_{x\sim P}[-\log q(x)]$

Start from the **discrete** definition of cross-entropy between a true distribution P and a model distribution Q:

$H(P,Q) = -\sum_x p(x)\log q(x)$.

Now recall the definition of an **expectation** under P:

$\mathbb{E}_{x\sim P}[f(x)] = \sum_x p(x)\, f(x)$.

If we choose the function $f(x) = -\log q(x)$, then plugging into the expectation definition gives:

$\mathbb{E}_{x\sim P}[-\log q(x)] = \sum_x p(x)\,(-\log q(x)) = -\sum_x p(x)\log q(x) = H(P,Q)$.

So we have shown: $H(P,Q) = \mathbb{E}_{x\sim P}[-\log q(x)]$.

Why this is useful for learning

We usually do not know $p(x)$, but we can sample data: $x_1, x_2, \ldots, x_n \sim P$.

Then the expectation can be approximated by a sample average (Monte Carlo / empirical risk):

$H(P,Q) = \mathbb{E}_{x\sim P}[-\log q(x)] \approx \frac{1}{n}\sum_{i=1}^n \big(-\log q(x_i)\big)$.

This is exactly the **negative log-likelihood loss** used to train probabilistic models and classifiers. This is directly estimable from data.KL divergence is not.

Why cross-entropy is the *right* practical objective

Cross-entropy has three crucial properties:

1. **Sample-based**
   It can be approximated using empirical averages.
2. **Differentiable**
   It provides clean gradients for learning.
3. **Equivalent to KL minimization**
   Minimizing cross-entropy automatically minimizes model mismatch.

Why this is actually desirable

Cross-entropy does **not** "ignore" data entropy but respects it.

It ensures that:

- learning focuses on model quality,
- unavoidable randomness is not confused with model error,
- gradients vanish only when the model truly matches the data.

10. Mutual Information

$$I(X;Y) = \mathrm{KL}\big(p(x,y)\|p(x)p(y)\big)$$

Equivalent form

$$I(X;Y) = \mathbb{E}_{p(x,y)}\left[\log\frac{p(x,y)}{p(x)p(y)}\right]$$

In [73]:

```python
def mutual_information_discrete(pxy):
    pxy = np.array(pxy, dtype=float)
    pxy = pxy / pxy.sum()
    px = pxy.sum(axis=1, keepdims=True)
    py = pxy.sum(axis=0, keepdims=True)
    mi = 0.0
    for i in range(pxy.shape[0]):
        for j in range(pxy.shape[1]):
            if pxy[i,j] > 0:
                mi += pxy[i,j] * np.log(pxy[i,j] / (px[i,0]*py[0,j]))
    return mi

pxy_indep = np.array([[0.25, 0.25],
                      [0.25, 0.25]])

pxy_dep = np.array([[0.40, 0.10],
                    [0.10, 0.40]])

mutual_information_discrete(pxy_indep), mutual_information_discrete(pxy_dep)
```

Out[73]:

```
(0.0, 0.19274475702175753)
```

Mutual Information (MI): A Simple Intuition

Mutual information measures **how much knowing one variable tells us about another**.

If knowing Y reduces our uncertainty about X, then X and Y have **positive mutual information**.

What MI really measures

- If X and Y are **independent**, then: $p(x,y) = p(x)p(y) \quad\Rightarrow\quad I(X;Y) = 0$
- If X and Y are **dependent**, then:
  - knowing Y gives information about X
  - mutual information is **positive**

A Simple 2×2 example

Let:

- $X \in \{0,1\}$
- $Y \in \{0,1\}$

Suppose the **joint distribution** is:

$X\backslash Y$	**0**	**1**
0	0.40	0.10
1	0.10	0.40

Step 1 — Marginal distributions

From the table:

$$p(X=0) = 0.5, \quad p(X=1) = 0.5$$
$$p(Y=0) = 0.5, \quad p(Y=1) = 0.5$$

If X and Y were independent, every entry would be 0.25, but they are not.

Step 2 — Intuition from the table

- When $X = 0$, Y is **more likely to be 0**
- When $X = 1$, Y is **more likely to be 1**

So:

> Knowing Y gives strong information about X, and vice versa.

This implies **high mutual information**.

Step 3 — Why MI is positive here

For example, at $(X = 0, Y = 0)$:

$$\frac{p(x,y)}{p(x)p(y)} = \frac{0.40}{0.25} > 1$$

Taking the log and averaging over all cells produces:

$$I(X;Y) > 0$$

Summary

> **Mutual information measures how far the joint distribution is from independence.**

- $I(X;Y) = 0$ → knowing one variable tells you nothing about the other
- Larger $I(X;Y)$ → stronger statistical dependence

11. Monte Carlo Estimation

Monte Carlo estimation is a general numerical technique for **approximating quantities using random samples**. It is especially useful when exact computation is difficult or impossible.

Core principle

If a quantity can be written as an **average** or an **integral**, then it can often be approximated by sampling.

Formally, for a function $f(x)$ and a random variable X with distribution P:

$$\mathbb{E}_{x\sim P}[f(x)] = \int f(x)\, p(x)\, dx$$

Monte Carlo estimation replaces this integral with a **sample average**: $\mathbb{E}_{x\sim P}[f(x)] \;\approx\; \frac{1}{N}\sum_{i=1}^N f(x_i), \quad x_i \sim P.$

Why it works

By the **Law of Large Numbers**, the average of many independent samples converges to the true expected value as the number of samples increases. No assumptions are needed about the shape or dimensionality of P.

What makes Monte Carlo powerful

- Works in **any dimension**
- Requires only the ability to **sample**
- Accuracy improves predictably as more samples are used
- Avoids explicit integration or summation

Trade-off

- Monte Carlo estimates are **random**
- Error decreases slowly, at rate: $\mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$
- High accuracy may require many samples

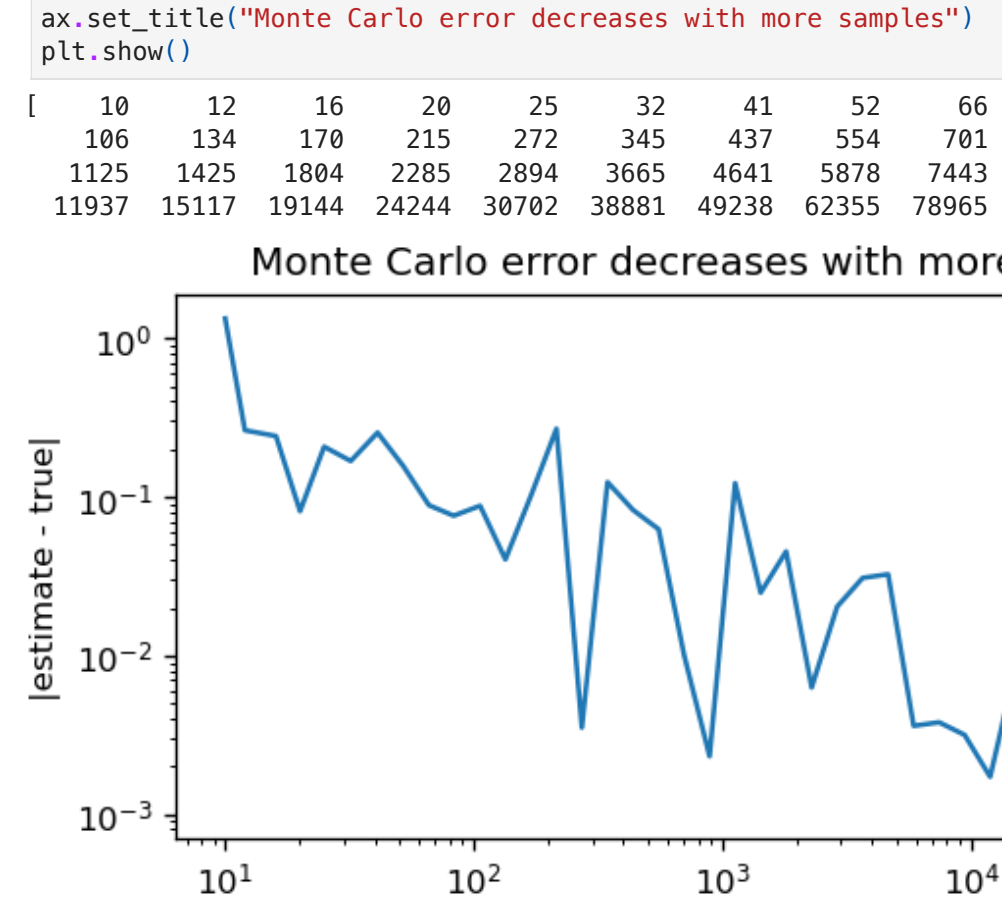
. Monte Carlo Estimation of Expectations

$$\mathbb{E}[f(X)] \approx \frac{1}{N}\sum_{i=1}^N f(x_i), \quad x_i \sim p(x)$$

In [75]:

```python
# Monte Carlo convergence: estimate E[X^2] for X~N(0,1) (true value = 1)
true = 1.0
Ns = np.unique(np.logspace(1, 5, 40).astype(int))
print(Ns)

errs = []
for N in Ns:
    X = np.random.normal(size=N)
    est = np.mean(X**2)
    errs.append(abs(est - true))

fig, ax = plt.subplots(figsize=(6,3))
ax.plot(Ns, errs)
ax.set_xscale("log")
ax.set_yscale("log")
ax.set_xlabel("N (samples)")
ax.set_ylabel("|estimate - true|")
ax.set_title("Monte Carlo error decreases with more samples")
plt.show()
```

```
[    10     12     16     20     25     32     41     52     66     83
    106    134    170    215    272    345    437    554    701    888
   1125   1425   1804   2285   2894   3665   4641   5878   7443   9426
  11937  15117  19144  24244  30702  38881  49238  62355  78965 100000]
```



12 Maximum Likelihood Estimation (MLE)

MLE picks the parameter values that make the observed data most likely under the assumed model.

We observe data points:

$$x_1, x_2, \ldots, x_n$$

We assume the data were generated i.i.d. from a distribution with parameters θ.

MLE solves:

$$\hat{\theta} = \arg\max_{\theta}\; \mathcal{L}(\theta)$$

where the **likelihood** is:

$$\mathcal{L}(\theta) = p(x_1, x_2, \ldots, x_n \mid \theta)$$

Because observations are i.i.d.:

$$p(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^n p(x_i \mid \theta)$$

Why we use **log-likelihood** instead of likelihood

In probabilistic modeling, we often want to fit a model $q_\theta(x)$ to data $x_1, x_2, \ldots, x_n \sim P$.

The most natural objective is the **likelihood**:

$$\mathcal{L}(\theta) = \prod_{i=1}^n q_\theta(x_i),$$

which measures how probable the observed data is under the model. For simplicity we rewrite $p(x_i \mid \theta)$ as $q_\theta(x_i)$.

So why don't we optimize this directly?

Reason 1 — Products of probabilities are numerically unstable

Probabilities are numbers **less than 1**.

Multiplying many of them quickly produces extremely small values: $0.1 \times 0.1 \times \cdots \times 0.1 \;\longrightarrow\; 0$

This causes:

- numerical underflow,
- loss of precision,
- unstable optimization.

Reason 2 — Logs turn products into sums

Taking the logarithm gives the **log-likelihood**:

$$\log\mathcal{L}(\theta) = \sum_{i=1}^n \log q_\theta(x_i).$$

This has major advantages:

- products → sums
- easier gradients
- numerically stable
- additive contributions per data point

Maximizing likelihood and maximizing log-likelihood are equivalent because $\log$ is monotonic.

Reason 3 — Log-likelihood matches expectations and learning theory

The average log-likelihood is: $\frac{1}{n}\sum_{i=1}^n \log q_\theta(x_i) \;\approx\; \mathbb{E}_{x\sim P}[\log q_\theta(x)]$.

Maximizing log-likelihood is therefore equivalent to **minimizing cross-entropy**: $H(P, Q_\theta) = \mathbb{E}_{x\sim P}[-\log q_\theta(x)]$.

This directly connects learning to:

- cross-entropy
- KL divergence
- information theory

Example 1: Bernoulli / Coin Toss MLE (θ = probability of Heads)

Model

Each toss is modeled as: $X \sim \mathrm{Bernoulli}(\theta)$

- Heads H (or $X = 1$) has probability θ
- Tails T (or $X = 0$) has probability $1 - \theta$

So for one observation:

- If $x = 1$: $p(x \mid \theta) = \theta$
- If $x = 0$: $p(x \mid \theta) = 1 - \theta$

Step-by-step likelihood for a specific sequence

Suppose we observe the **exact sequence**: $H,\ T,\ H,\ H,\ T$

Write the probability for each outcome explicitly:

- $P(H) = \theta$
- $P(T) = 1 - \theta$
- $P(H) = \theta$
- $P(H) = \theta$
- $P(T) = 1 - \theta$

Because tosses are independent, multiply them:

$\mathcal{L}(\theta) = \theta \times (1-\theta) \times \theta \times \theta \times (1-\theta)$

Group terms:

$= \underbrace{\theta \times \theta \times \theta}_{\text{3 heads}} \times \underbrace{(1-\theta) \times (1-\theta)}_{\text{2 tails}}$

So:

$\mathcal{L}(\theta) = \theta^3(1-\theta)^2$

General form (k heads out of n tosses)

If we observe k heads and $n-k$ tails, then the likelihood is:

$\mathcal{L}(\theta) = \underbrace{\theta \times \cdots \times \theta}_{k\text{ times}} \times \underbrace{(1-\theta) \times \cdots \times (1-\theta)}_{n-k\text{ times}} = \theta^k(1-\theta)^{n-k}$

Maximizing the likelihood (using log-likelihood)

Take logs to convert products into sums:

$\ell(\theta) = \log\mathcal{L}(\theta) = \log\big(\theta^k(1-\theta)^{n-k}\big)$

Use log rules (split product into sum):

$\ell(\theta) = \log(\theta^k) + \log((1-\theta)^{n-k})$

Bring down powers:

$\ell(\theta) = k\log\theta + (n-k)\log(1-\theta)$

Differentiate and set to zero:

$\frac{d\ell}{d\theta} = \frac{k}{\theta} - \frac{n-k}{1-\theta} = 0$

Solve step by step:

1. Move one term to the other side:

$\frac{k}{\theta} = \frac{n-k}{1-\theta}$

2. Cross-multiply:

$k(1-\theta) = (n-k)\theta$

3. Expand left side:

$k - k\theta = (n-k)\theta$

4. Expand right side:

$k - k\theta = n\theta - k\theta$

5. Add $k\theta$ to both sides:

$k = n\theta$

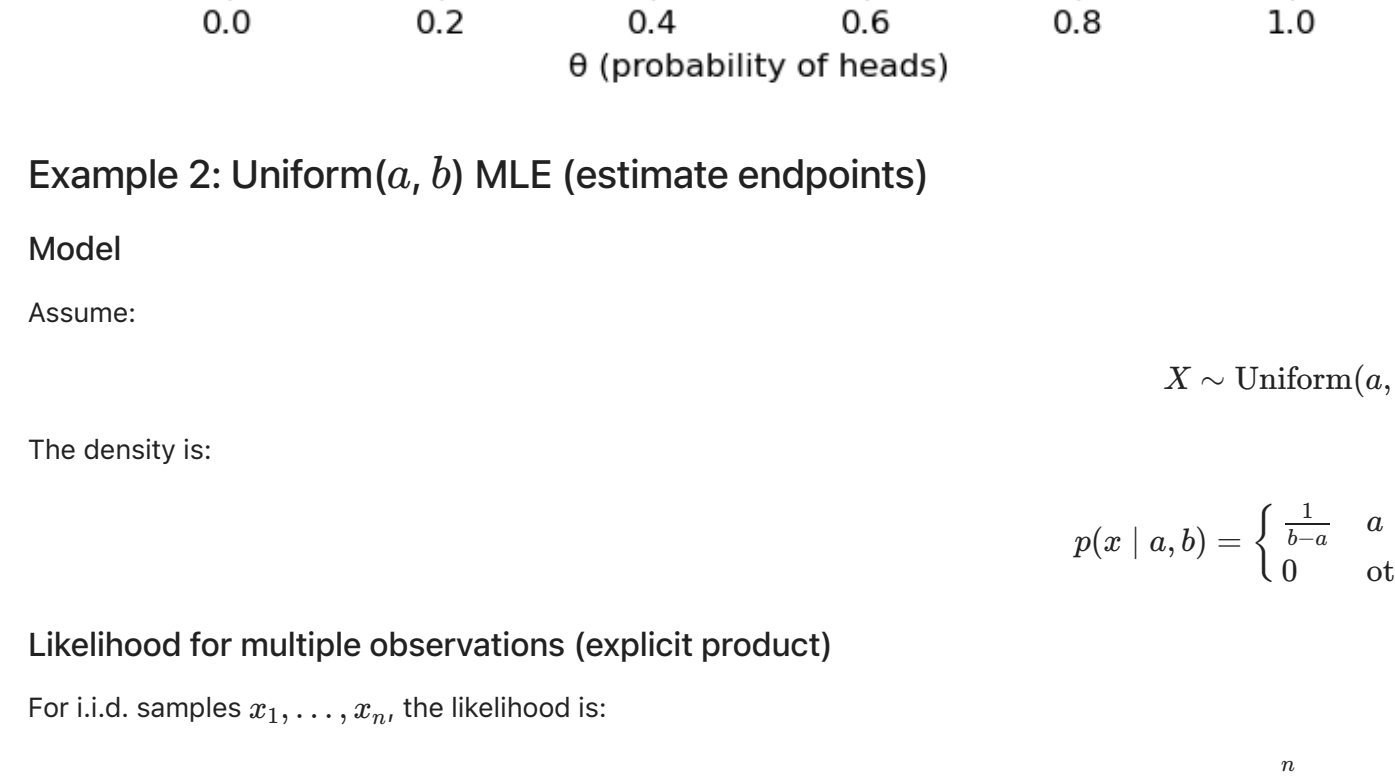
6. Divide by n:

$\boxed{\hat{\theta}_{MLE} = \frac{k}{n}}$

This matches intuition: **MLE equals the empirical frequency of heads**.

In [76]:

```python
# Visualizing the Bernoulli likelihood for an example:
# Suppose n=10 tosses and k=8 heads
n = 10
k = 2

theta = np.linspace(0.001, 0.999, 1000)
L = theta**k * (1-theta)**(n-k)

plt.figure()
plt.plot(theta, L)
plt.axvline(k/n, linestyle="--", label=f"MLE = k/n = {k/n:.2f}")
plt.xlabel("θ (probability of heads)")
plt.ylabel("Likelihood L(θ)")
plt.title("Bernoulli Likelihood")
plt.legend()
plt.show()
```



Example 2: Uniform(a, b) MLE (estimate endpoints)

Model

Assume:

$$X \sim \mathrm{Uniform}(a,b)$$

The density is:

$$p(x \mid a,b) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

Likelihood for multiple observations (explicit product)

For i.i.d. samples $x_1, \ldots, x_n$, the likelihood is:

$$\mathcal{L}(a,b) = \prod_{i=1}^n p(x_i \mid a,b)$$

If **every** x_i lies inside $[a,b]$, then each term equals $\frac{1}{b-a}$:

$$\mathcal{L}(a,b) = \underbrace{\frac{1}{b-a} \times \frac{1}{b-a} \times \cdots \times \frac{1}{b-a}}_{n\text{ times}} = (b-a)^{-n}$$

If **any** x_i is outside $[a,b]$, then at least one term is 0, so:

$$\mathcal{L}(a,b) = 0$$

So we can write:

$$\mathcal{L}(a,b) = \begin{cases} (b-a)^{-n} & a \le \min(x_i) \text{ and } b \ge \max(x_i) \\ 0 & \text{otherwise} \end{cases}$$

How to maximize it?

When the interval covers the data, likelihood is $(b-a)^{-n}$.

Maximizing $(b-a)^{-n}$ is the same as **minimizing** $(b-a)$ (negative exponent).

So we want the **tightest interval** that still contains all data:

$$\boxed{\hat{a} = \min(x_i), \quad \hat{b} = \max(x_i)}$$

In [77]:

```python
np.random.seed(0)
data = np.random.uniform(0, 1, size=20)

a_hat = data.min()
b_hat = data.max()

a_hat, b_hat
```

Out[77]:

```
(0.02021839744032572, 0.9636627605010293)
```

In [78]:

```python
# Numerical search for (a,b) without using the analytic formula
def logL_uniform(a, b, x):
    if a <= x.min() and b >= x.max() and b > a:
        return -len(x) * np.log(b - a)
    return -np.inf

a_grid = np.linspace(-0.2, 0.8, 250)
b_grid = np.linspace(0.2, 1.4, 250)

best = (-np.inf, None)
for a in a_grid:
    for b in b_grid:
        ll = logL_uniform(a, b, data)
        if ll > best[0]:
            best = (ll, (a, b))

best_ll, (a_num, b_num) = best
(a_hat, b_hat), (a_num, b_num), best_ll
```

Out[78]:

```
((0.02021839744032572, 0.9636627605010293),
 (0.016867469879518038, 0.9662650602409637),
 1.0385522186552922)
```

Example 3: Gaussian (Normal) MLE for Mean and Variance

Model

Assume:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

The density for one observation is:

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Step 1: Write the likelihood as a product (i.i.d.)

For data $x_1, \ldots, x_n$:

$$\mathcal{L}(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

Write the first few terms explicitly:

$$\mathcal{L}(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_1-\mu)^2}{2\sigma^2}\right)\right) \times \left(\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_2-\mu)^2}{2\sigma^2}\right)\right) \times \cdots \times \left(\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_n-\mu)^2}{2\sigma^2}\right)\right)$$

Now group constants and exponentials.

Constants

The constant factor $\frac{1}{\sqrt{2\pi\sigma^2}}$ appears n times, so:

$$\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n$$

Exponentials

A product of exponentials becomes the exponential of a sum:

$$\prod_{i=1}^n \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right) = \exp\left(-\sum_{i=1}^n \frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

So:

$$\mathcal{L}(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\sum_{i=1}^n \frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

Step 2: Take log-likelihood

Define $\ell(\mu,\sigma^2) = \log \mathcal{L}(\mu,\sigma^2)$.

Apply log rules:

$$\ell(\mu,\sigma^2) = \log\left(\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n\right) + \log\left(\exp\left(-\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}\right)\right)$$

Simplify each term:

1. First term (bring down the power):

$$\log\left(\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n\right) = n\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) = -\frac{n}{2}\log(2\pi\sigma^2)$$

2. Second term (log(exp(.)) cancels):

$$\log(\exp(\cdot)) = \cdot \Rightarrow -\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}$$

Therefore:

$$\boxed{\ell(\mu,\sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

Step 3: MLE for μ

Only the second term depends on μ. Let:

$$S(\mu) = \sum_{i=1}^{n}(x_i-\mu)^2$$

Then:

$$\ell(\mu,\sigma^2) = C - \frac{1}{2\sigma^2}S(\mu)$$

Differentiate w.r.t. μ. For one term:

$$\frac{d}{d\mu}(x_i-\mu)^2 = 2(x_i-\mu)(-1) = -2(x_i-\mu)$$

So:

$$\frac{\partial\ell}{\partial\mu} = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(-2(x_i-\mu)) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)$$

Set equal to zero:

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0 \Rightarrow \sum_{i=1}^{n}(x_i-\mu) = 0$$

Expand:

$$\sum_{i=1}^{n}x_i - \sum_{i=1}^{n}\mu = 0$$

But $\sum_{i=1}^{n}\mu = n\mu$, so:

$$\sum_{i=1}^{n}x_i - n\mu = 0 \Rightarrow n\mu = \sum_{i=1}^{n}x_i$$

Divide by n:

$$\boxed{\hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}x_i}$$

Step 4: MLE for σ^2

Start from:

$$\ell(\mu,\sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Let $S = \sum_{i=1}^{n}(x_i-\mu)^2$ (treat μ as fixed when differentiating w.r.t. σ^2).

Differentiate term by term.

Term A

$$\frac{d}{d\sigma^2}\left(-\frac{n}{2}\log(2\pi\sigma^2)\right) = -\frac{n}{2}\cdot\frac{1}{\sigma^2}$$

Term B

Rewrite:

$$-\frac{1}{2\sigma^2}S = -\frac{S}{2}(\sigma^2)^{-1}$$

Derivative of $(\sigma^2)^{-1}$ is $-(\sigma^2)^{-2}$, so:

$$\frac{d}{d\sigma^2}\left(-\frac{S}{2}(\sigma^2)^{-1}\right) = -\frac{S}{2}\cdot(-(\sigma^2)^{-2}) = \frac{S}{2(\sigma^2)^2}$$

Combine

$$\frac{\partial\ell}{\partial\sigma^2} = -\frac{n}{2}\cdot\frac{1}{\sigma^2} + \frac{S}{2(\sigma^2)^2}$$

Set to zero:

$$-\frac{n}{2}\cdot\frac{1}{\sigma^2} + \frac{S}{2(\sigma^2)^2} = 0$$

Multiply both sides by $2(\sigma^2)^2$:

$$-n\sigma^2 + S = 0 \Rightarrow S = n\sigma^2$$

So:

$$\boxed{\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2}$$

Note: This uses $1/n$ (MLE). The common $1/(n-1)$ form is the **unbiased** estimator.

13 Summary of Terms

This table summarizes the most important quantities and how to interpret them.

Quantity	Math (discrete)	Intuition
Probability of one outcome	$p(x)$	How likely an event is
Log-probability	$\log p(x)$	Additive "score"; relates to information
Likelihood	$\mathcal{L}(\theta) = \prod_i p_\theta(x_i)$	How plausible parameters θ are given data
Log-likelihood	$\ell(\theta) = \sum_i \log p_\theta(x_i)$	Sum of evidence / negative surprise
NLL loss	$\mathrm{NLL}(\theta) = -\ell(\theta)$	What we minimize in ML training
Entropy	$H(p) = -\sum_x p(x)\log p(x)$	Inherent uncertainty in p
Cross-entropy	$H(p,q) = -\sum_x p(x)\log q(x)$	Surprise when data from p is modeled as q
KL-divergence	$\mathrm{KL}(p\|q) = \sum_x p(x)\log\frac{p(x)}{q(x)}$	Extra surprise / inefficiency from using q instead of p
Key identity	$H(p,q) = H(p) + \mathrm{KL}(p\|q)$	Cross-entropy = uncertainty + mismatch penalty

14 Generative models: generation vs. sampling

A **generative model** answers two distinct questions:

1. **Generation (learning)**
   What probability distribution should we use to describe the data?
2. **Sampling (usage)**
   Once we have that distribution, how do we generate new data points from it?

These are conceptually different phases.

Starting point: a simple discrete data table

Suppose our observed data consists of pairs (X,Y) with joint distribution:

$X\backslash Y$	0	1
1	0.20	0.05
2	0.25	0.10
3	0.10	0.25
4	0.05	0.00

This table *is the data distribution* $p_{data}(x,y)$.

Phase 1 — Generation (learning the distribution)

In the **generation phase**, the model's goal is to **learn or define** a probability distribution that matches the data.

Here, the generative model is simply:

$$q(x,y) \approx p_{\mathrm{data}}(x,y)$$

For this toy example, the model could literally **store the table** or estimate it from counts.

This phase answers:

- Which outcomes are likely?
- Which outcomes are rare?
- How are X and Y statistically related?

Nothing is being generated yet, we are only defining a probability law.

Phase 2 — Sampling (using the model)

Once the distribution $q(x,y)$ is defined, we can **sample new data** from it.

Sampling means:

1. Randomly draw a pair (x,y) according to the probabilities in the table
2. Treat the sampled pair as a **new synthetic data point**

For example, one possible sample sequence might be:

$$(2,0),\ (3,1),\ (1,0),\ (3,1),\ \ldots$$

Each draw respects the learned probabilities.

Conditional sampling

Because the model defines a **joint distribution**, we can also sample conditionally.

Example: sample X given $Y=1$

From the table:

$$p(x \mid Y=1) = \frac{p(x,1)}{p(Y=1)}$$

This allows us to:

- fix $Y=1$,
- sample X according to the conditional distribution.

This is the simplest form of **controlled generation**.

Why this distinction matters

Phase	What happens	Typical ML language
Generation	Learn $q(x)$ or $q(x,y)$	training, fitting, estimating
Sampling	Draw new data from q	generation, synthesis

Many modern models follow this exact split:

- **VAEs**: learn parameters → sample latent variables → decode
- **Diffusion models**: learn denoising distribution → sample by reverse process
- **Autoregressive models**: learn conditional probabilities → sample sequentially

Summary

> **Generation defines the rules of the random world.**
> **Sampling lets you play inside that world.**

This simple table already contains *everything* a generative model needs — modern models just scale this idea to high dimensions.

15 Connecting generation & sampling to MLE and cross-entropy training

We now connect the two phases of a generative model to **how models are actually trained** in practice.

Phase 1 revisited — Generation = learning a distribution

Recall that in the **generation (learning) phase**, the goal is to learn a model distribution:

$$q_\theta(x,y)$$

that matches the true (unknown) data distribution:

$$p_{\mathrm{data}}(x,y).$$

We do **not** know p_{data} explicitly.
We only observe samples:

$$(x_1,y_1),(x_2,y_2),\ldots,(x_n,y_n).$$

So the question becomes:

> *How should we choose θ so that q_θ explains the observed data well?*

Maximum Likelihood Estimation (MLE)

The most natural principle is **maximum likelihood**.

We choose θ to maximize the probability that the model assigns to the observed data:

$$\theta^* = \arg\max_\theta \prod_{i=1}^{n} q_\theta(x_i,y_i).$$

This says:

> *A good generative model should assign high probability to the data we actually see.*

From likelihood to log-likelihood

Because products of probabilities are numerically unstable, we maximize the **log-likelihood** instead:

$$\arg\max_\theta \sum_{i=1}^{n}\log q_\theta(x_i,y_i).$$

This does **not change the optimum**, but makes optimization:

- stable,
- additive across samples,
- easy to differentiate.

From log-likelihood to cross-entropy

Dividing by n, the average log-likelihood becomes:

$$\frac{1}{n}\sum_{i=1}^{n}\log q_\theta(x_i,y_i).$$

As n grows, this empirical average converges to an expectation under the data distribution:

$$\mathbb{E}_{(x,y)\sim p_{\mathrm{data}}}\big[\log q_\theta(x,y)\big].$$

Maximizing log-likelihood is therefore equivalent to minimizing:

$$\mathbb{E}_{(x,y)\sim p_{\mathrm{data}}}\big[-\log q_\theta(x,y)\big],$$

which is exactly the **cross-entropy**:

$$H\big(p_{\mathrm{data}}, q_\theta\big).$$

Why cross-entropy is the right training objective

Recall the identity:

$$H(P,Q) = H(P) + \mathrm{KL}(P\|Q).$$

Here:

- $P = p_{\mathrm{data}}$ (fixed, unknown)
- $Q = q_\theta$ (learnable model)

Since $H(P)$ **does not depend on** θ, minimizing cross-entropy is equivalent to:

$$\min_\theta \mathrm{KL}(p_{\mathrm{data}}\|q_\theta).$$

So training via cross-entropy means:

> **Make the model distribution as close as possible to the data distribution.**

Phase 2 revisited — Sampling after training

Once training is complete, we have a learned distribution:

$$q_{\theta^*}(x,y).$$

Now we can **sample from it**:

- generate new (x,y) pairs,
- sample conditionally (e.g., $x \mid y=1$),
- or marginalize as needed.

Most importantly:

- **MLE / cross-entropy only affects the learning phase**
- **Sampling uses the learned distribution, unchanged**

Summary

Concept	What it does
MLE	Chooses the best model parameters
Log-likelihood	Makes optimization stable
Cross-entropy	Turns learning into expectation minimization
KL divergence	Measures model–data mismatch
Sampling	Uses the learned distribution to generate data

A simple 1D generative example

We assume data is generated by a **known function with noise**:

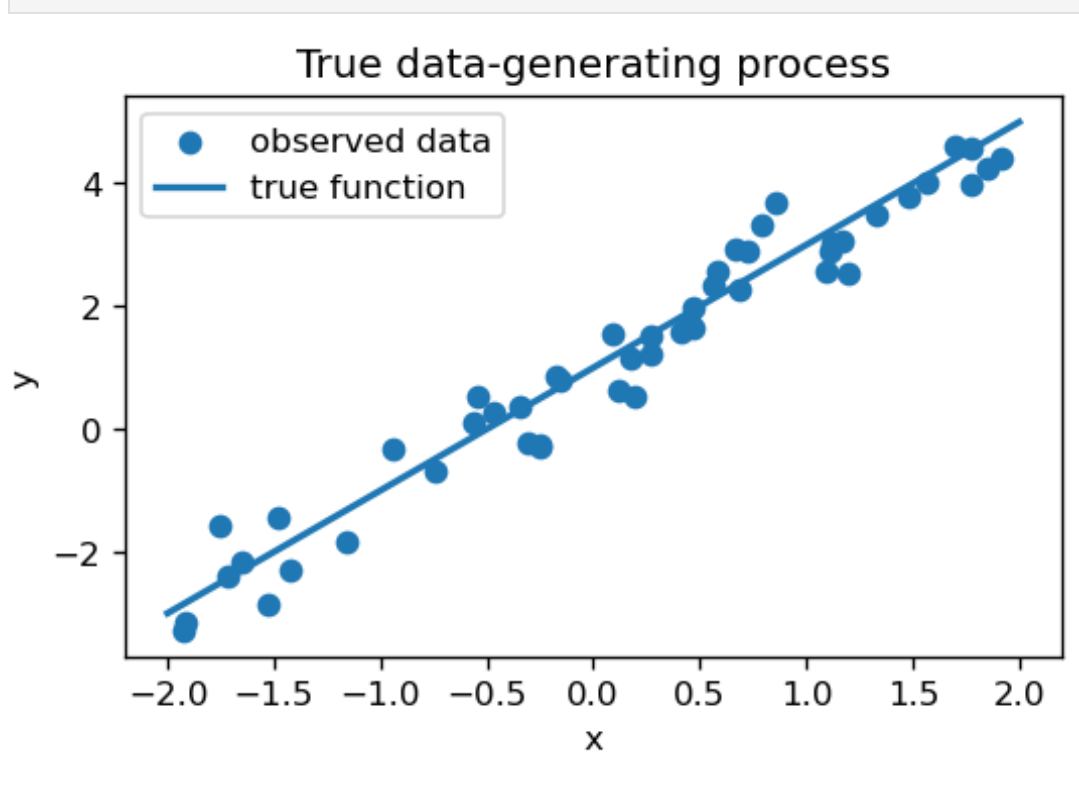
$y = f(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,\sigma^2)$

Here:

- x is the input (1D)
- y is the output (1D)
- $f(x)$ is the underlying structure
- randomness comes from noise

This already defines a **generative process**.

```
In [79]: import numpy as np
         import matplotlib.pyplot as plt

         np.random.seed(0)

         # True function
         def f_true(x):
             return 2*x + 1

         # Generate training data
         N = 50
         x = np.random.uniform(-2, 2, size=N)
         noise = np.random.normal(0, 0.5, size=N)
         y = f_true(x) + noise

         # Plot data
         xs = np.linspace(-2, 2, 200)

         plt.figure(figsize=(5,3))
         plt.scatter(x, y, label="observed data")
         plt.plot(xs, f_true(xs), label="true function", linewidth=2)
         plt.xlabel("x")
         plt.ylabel("y")
         plt.title("True data-generating process")
         plt.legend()
         plt.show()
```



Generation phase: learning a probabilistic model using MLE

In this step, we **learn the parameters of a generative model** from data.

What model are we assuming?

We assume that the data is generated by a **linear function with Gaussian noise**:

$$y = ax + b + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,\sigma^2)$$

This defines a **conditional probability distribution**:

$$p(y \mid x; a,b) = \mathcal{N}\big(y \mid ax+b,\ \sigma^2\big)$$

Our goal in the generation (training) phase is to find the parameters a and b that make the observed data **most likely** under this model.

Why Maximum Likelihood Estimation (MLE)?

Under the Gaussian noise assumption, the **log-likelihood** of the data is:

$$\log \mathcal{L}(a,b) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(y_i - (ax_i+b)\big)^2 \ +\ \text{constant}$$

Maximizing this log-likelihood is therefore **equivalent to minimizing the sum of squared errors**.

This is exactly the **least-squares problem**.

So:

> **Least squares = Maximum Likelihood Estimation**
> *(when noise is Gaussian)*

What the code does (step by step)

1. Builds a **design matrix** containing the inputs x and a bias term:

$$A = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}$$

2. Solves the least-squares problem:

$$\min_{a,b}\sum_{i=1}^{n}(y_i - (ax_i+b))^2$$

3. Obtains the MLE estimates $\hat{a}$ and $\hat{b}$.

4. Defines the learned model:

$$f_{\mathrm{model}}(x) = \hat{a}x + \hat{b}$$

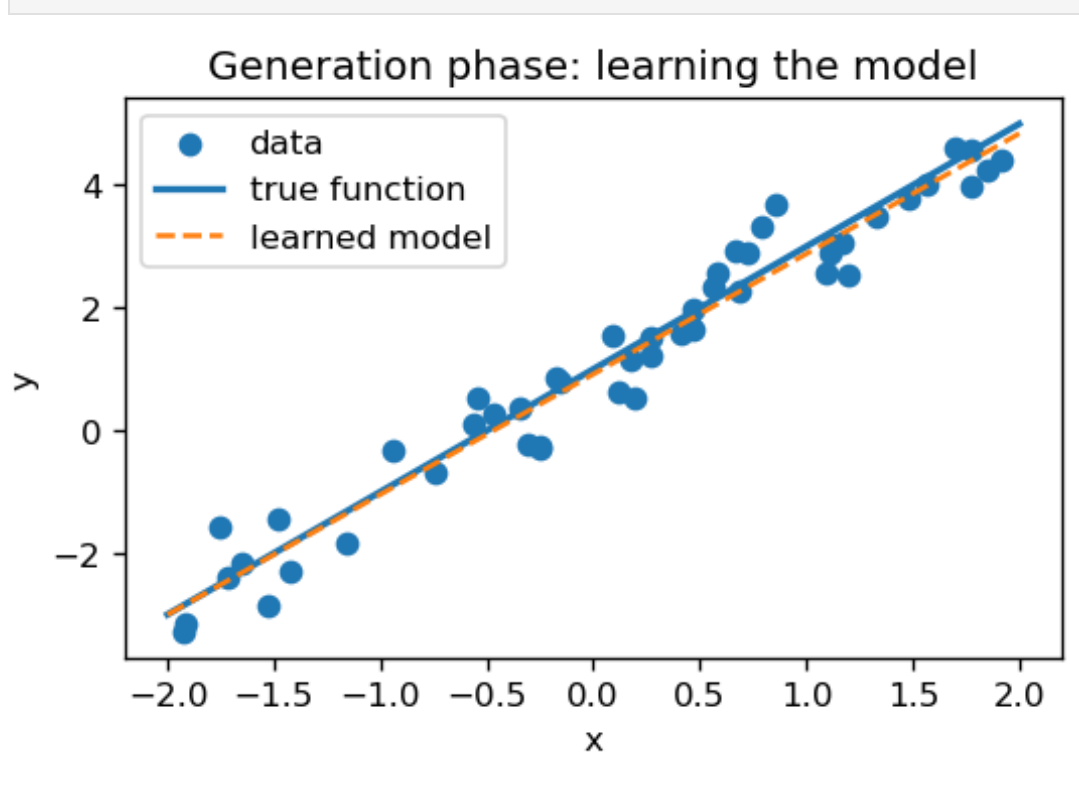
5. Plots:
   - the observed data,
   - the true (unknown) generating function,
   - the learned model.

Why this belongs in a *generative* model discussion

Although this looks like simple regression, it is already a **generative model**:

- We learn a distribution $p(y \mid x)$
- We learn its parameters using MLE
- We can later **sample** new y values given x

```
In [80]: # Fit linear model y = ax + b using least squares (MLE)
         A = np.vstack([x, np.ones_like(x)]).T
         # do the least squares solution
         a_hat, b_hat = np.linalg.lstsq(A, y, rcond=None)[0]

         def f_model(x):
             return a_hat*x + b_hat

         plt.figure(figsize=(5,3))
         plt.scatter(x, y, label="data")
         plt.plot(xs, f_true(xs), label="true function", linewidth=2)
         plt.plot(xs, f_model(xs), label="learned model", linestyle="--")
         plt.xlabel("x")
         plt.ylabel("y")
         plt.title("Generation phase: learning the model")
         plt.legend()
         plt.show()

         a_hat, b_hat
```



```
Out[80]: (1.9615916184136912, 0.9195781947622427)
```

Sampling phase: generating new data from the learned model

After MLE, we have learned a **probabilistic model**, not just a curve.

From the generation (training) phase we obtained: $q(y \mid x) = \mathcal{N}\Big(y \,\Big|\, \hat{a}x+\hat{b},\ \hat{\sigma}^2\Big)$

This means:

- $\hat{a}x+\hat{b}$ is the **mean** of the distribution,
- $\hat{\sigma}^2$ captures the **noise level** in the data.

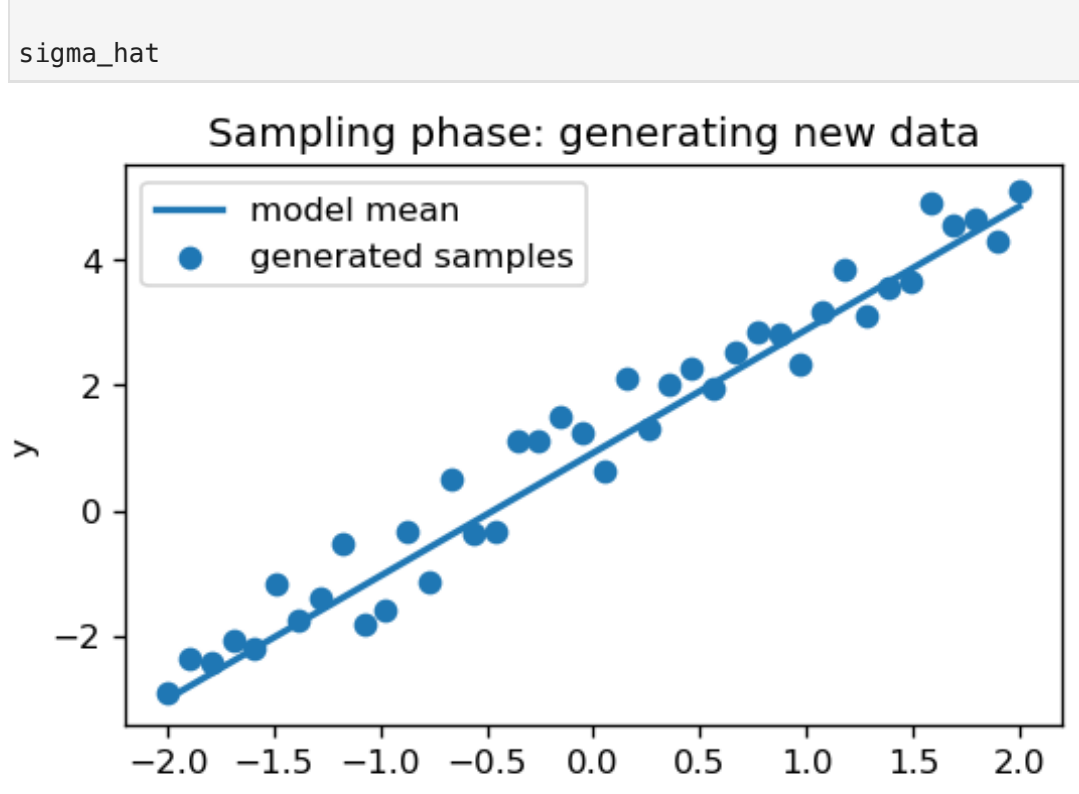
In the **sampling phase**, we:

1. choose new input values x,
2. sample noise $\varepsilon \sim \mathcal{N}(0,\hat{\sigma}^2)$,
3. generate synthetic outputs:

$$y = \hat{a}x + \hat{b} + \varepsilon.$$

This produces *new data points* that follow the same statistical rules as the original data

```
In [81]: # Estimate noise variance
         residuals = y - f_model(x)
         sigma_hat = np.std(residuals)

         # Sample new data
         x_new = np.linspace(-2, 2, 40)
         y_sampled = f_model(x_new) + np.random.normal(0, sigma_hat, size=len(x_new))

         plt.figure(figsize=(5,3))
         plt.plot(xs, f_model(xs), label="model mean", linewidth=2)
         plt.scatter(x_new, y_sampled, label="generated samples")
         plt.xlabel("x")
         plt.ylabel("y")
         plt.title("Sampling phase: generating new data")
         plt.legend()
         plt.show()

         sigma_hat
```



```
Out[81]: 0.46247366646128796
```

Why Mean Squared Error (MSE) equals Gaussian log-likelihood

This equivalence explains why **MSE is not arbitrary** but it is a probabilistic objective.

Step 1 — Assume a probabilistic model

Suppose the data is generated as:

$$y = f_\theta(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,\sigma^2)$$

This defines a **conditional distribution**:

$$p(y \mid x;\theta) = \mathcal{N}\big(y \mid f_\theta(x),\ \sigma^2\big)$$

Step 2 — Write the Gaussian likelihood

For a single data point (x_i,y_i):

$$p(y_i \mid x_i;\theta) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(y_i - f_\theta(x_i))^2}{2\sigma^2}\right)$$

For n independent samples, the likelihood is the product:

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} p(y_i \mid x_i;\theta)$$

Step 3 — Take the log-likelihood

Taking logs turns the product into a sum:

$$\log\mathcal{L}(\theta) = \sum_{i=1}^{n}\log p(y_i \mid x_i;\theta)$$

Substituting the Gaussian form:

$$\log\mathcal{L}(\theta) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - f_\theta(x_i))^2$$

Step 4 — Identify what depends on the model

The first term:

$$-\frac{n}{2}\log(2\pi\sigma^2)$$

does **not** depend on θ.

The only term that depends on the model is:

$$\sum_{i=1}^{n}(y_i - f_\theta(x_i))^2$$

Step 5 — Maximizing likelihood = minimizing MSE

Maximizing the log-likelihood is therefore equivalent to minimizing:

$$\sum_{i=1}^{n}(y_i - f_\theta(x_i))^2$$

Dividing by n gives the **Mean Squared Error**:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - f_\theta(x_i))^2$$

Final conclusion (important)

> **MSE is the negative log-likelihood of a Gaussian noise model (up to a constant).**

So when you minimize MSE, you are implicitly assuming:

- Gaussian noise
- constant variance
- independent errors

Summary

Loss	Implied noise model
MSE	Gaussian
MAE	Laplace
Huber	Gaussian + Laplace
Cross-entropy	Categorical / Bernoulli

Loss functions are **probabilistic assumptions in disguise**.

> **Using MSE means you believe your errors are Gaussian — whether you say so or not.**

Use of Generative AI

Portions of this material were developed with the assistance of **generative artificial intelligence tools.
The author reviewed, edited, and validated all content, including explanations, code, and examples, and assumes full responsibility for accuracy and interpretation.